The M-CATs

SEMINAR 17

Table of Contents

CH 9
Organic Chemistry
Carboxylic Acid Derivatives

CH 10
Organic Chemistry
Nitrogen & Phosphorus

CH 12
Organic Chemistry
Separation & Purification

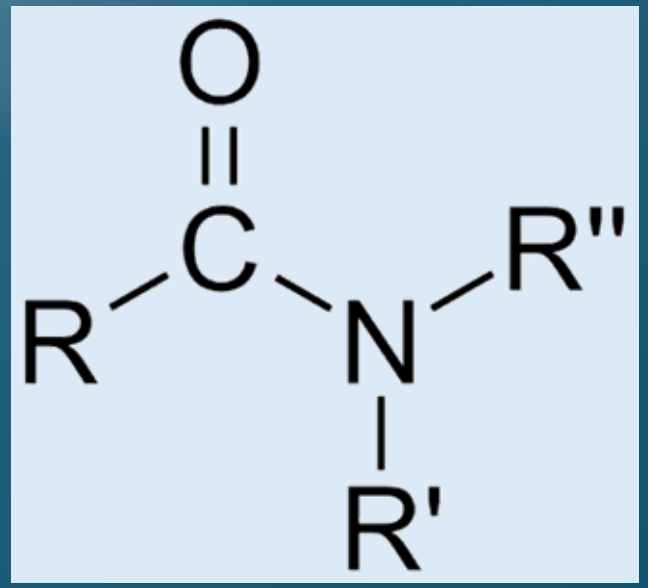


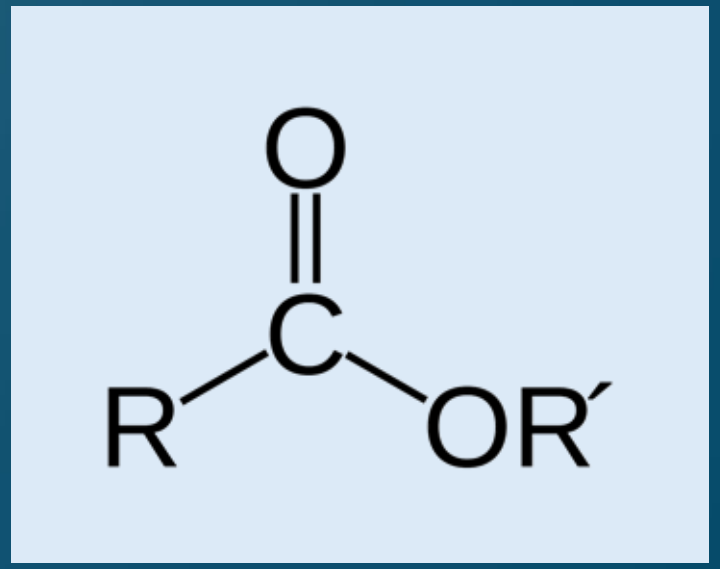
Organic Chemistry Chapter 9

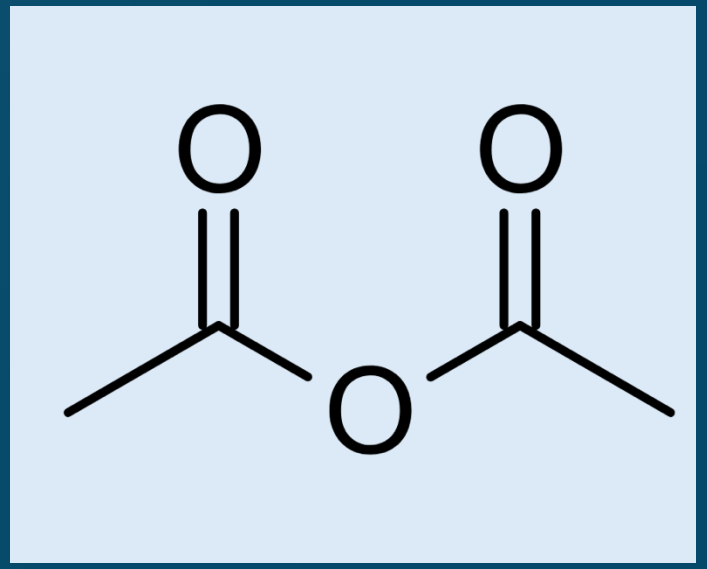
Carboxylic Acid Derivatives



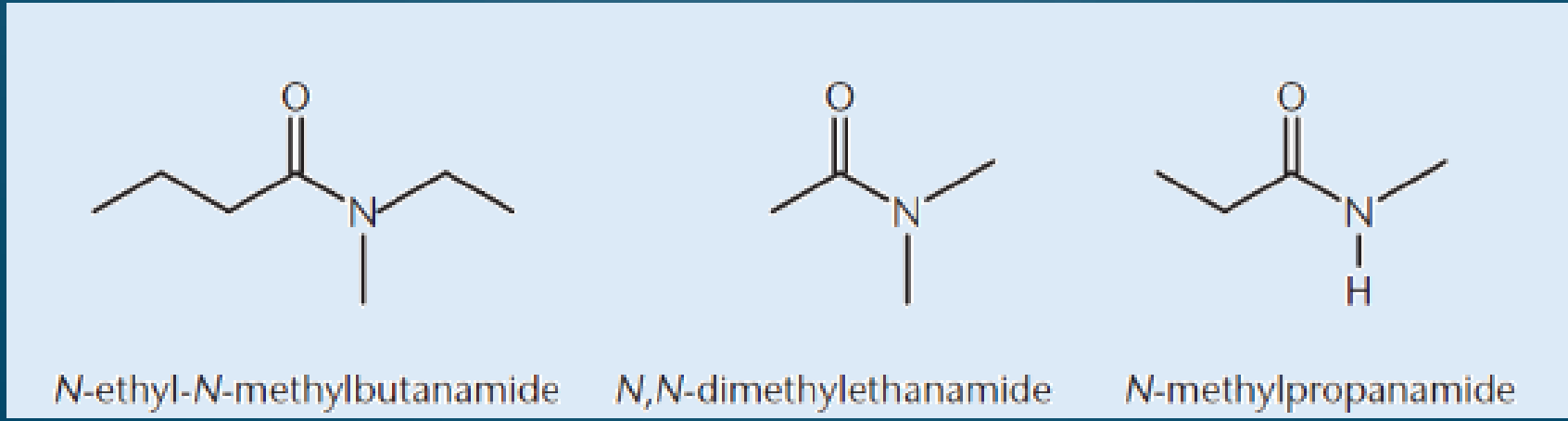
9.1. Amides, Esters, Anhydrides

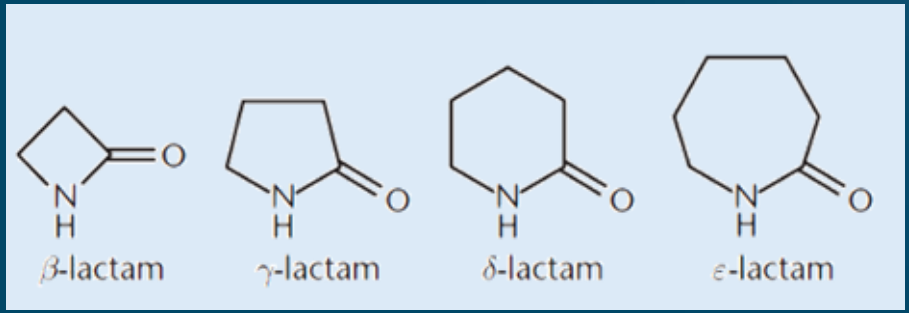


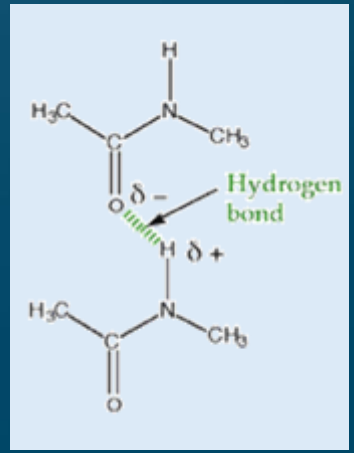




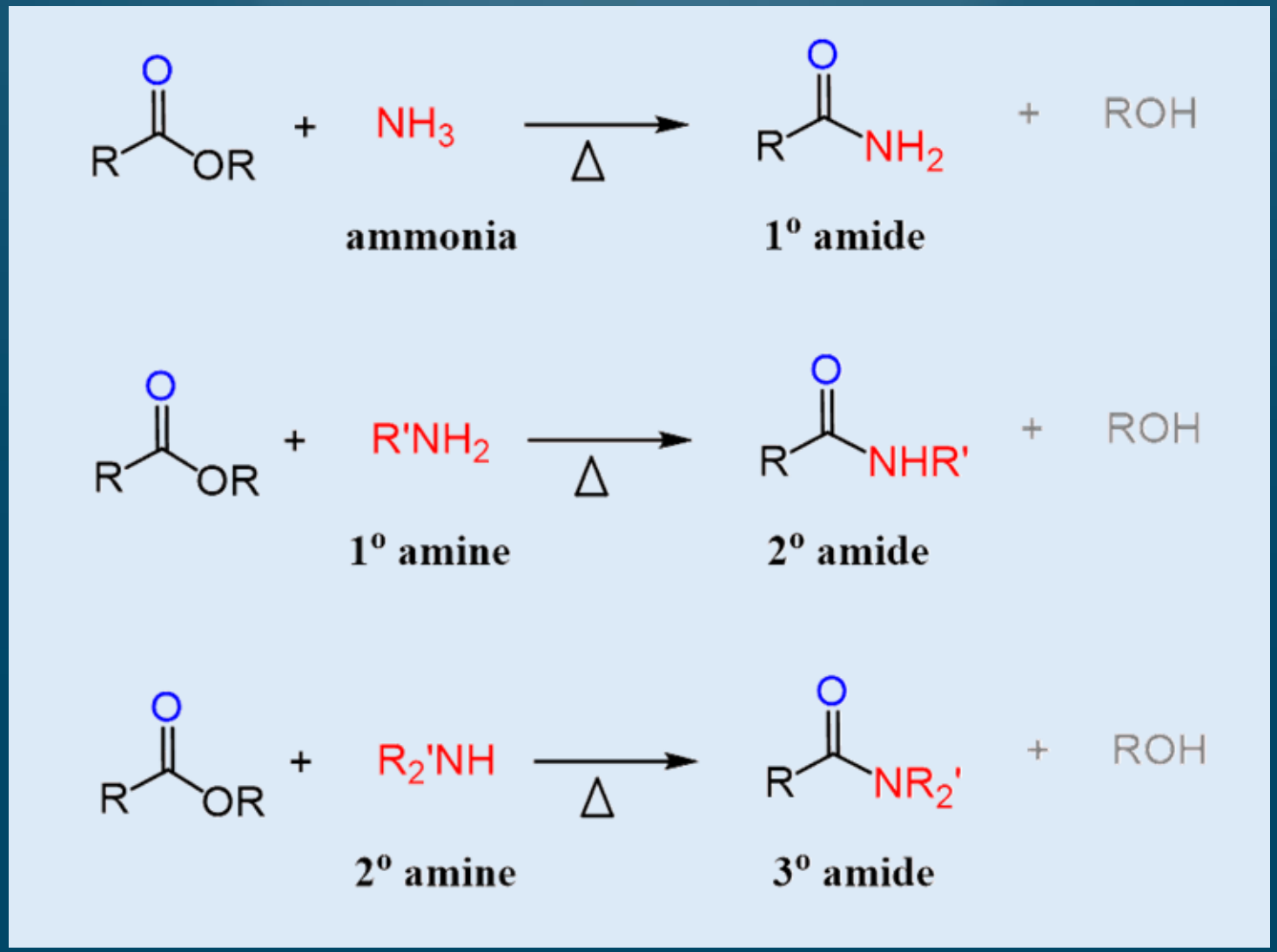
Amides: Nomenclature & Properties



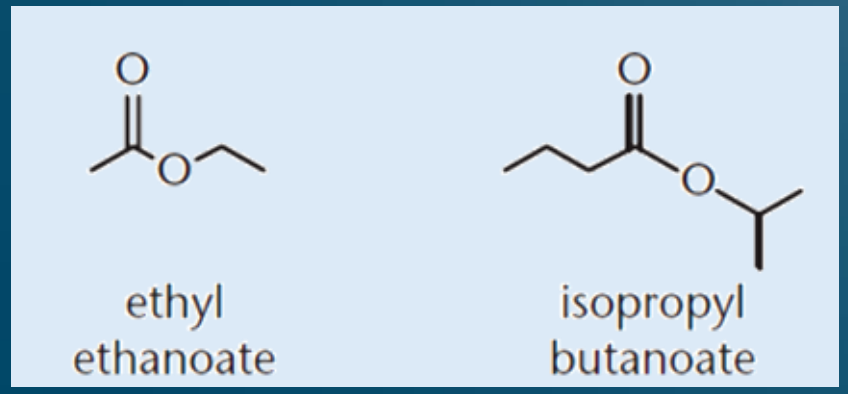


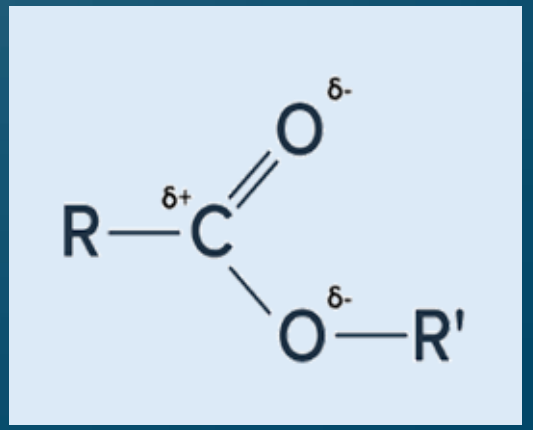


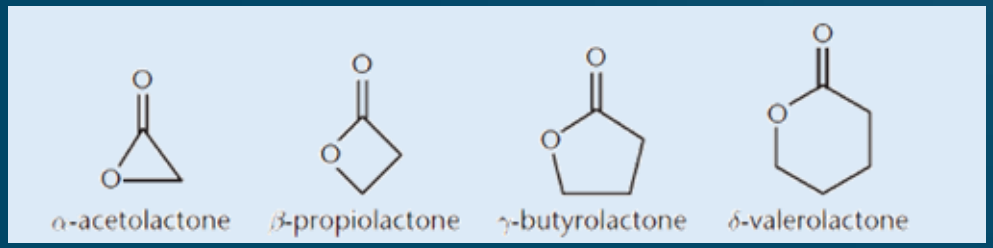
Amides: Formation



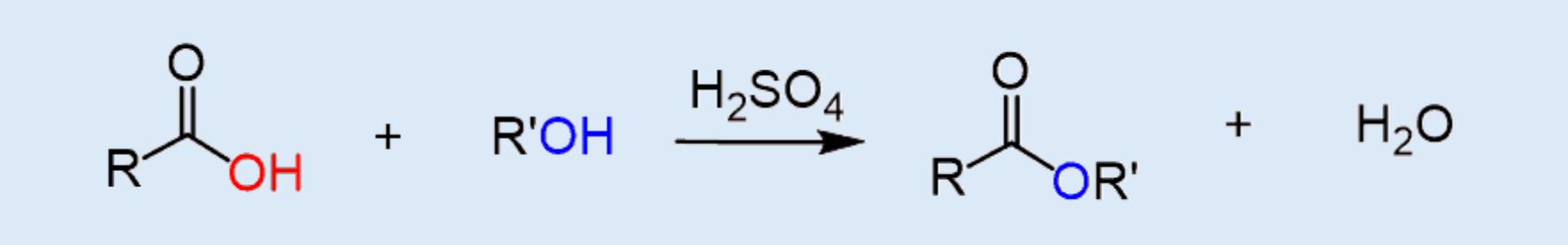
Esters: Nomenclature & Properties

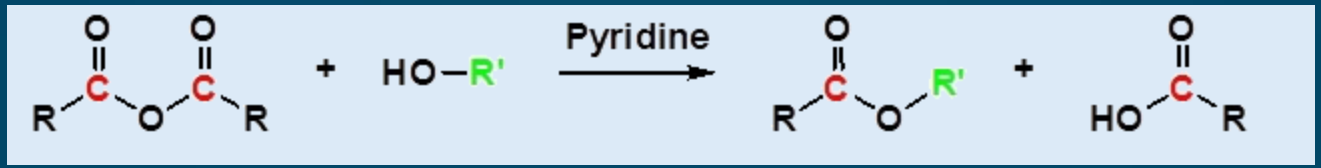




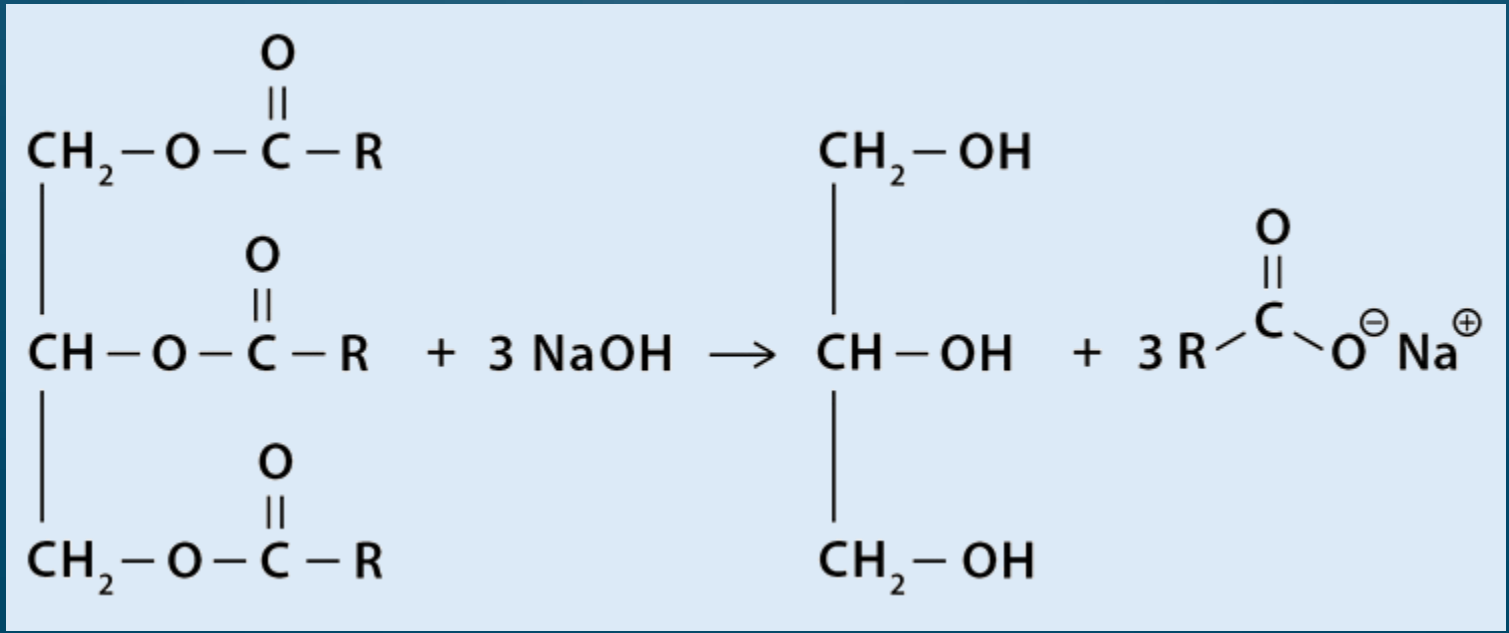


Esters: Formation

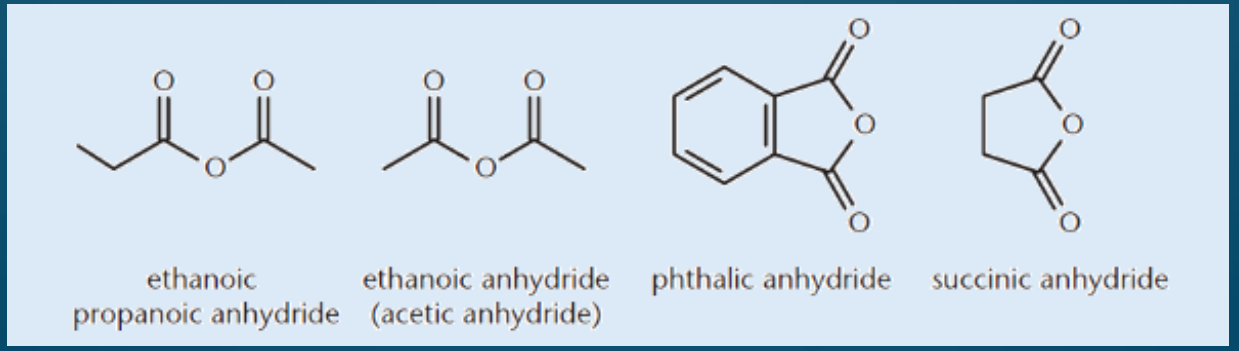




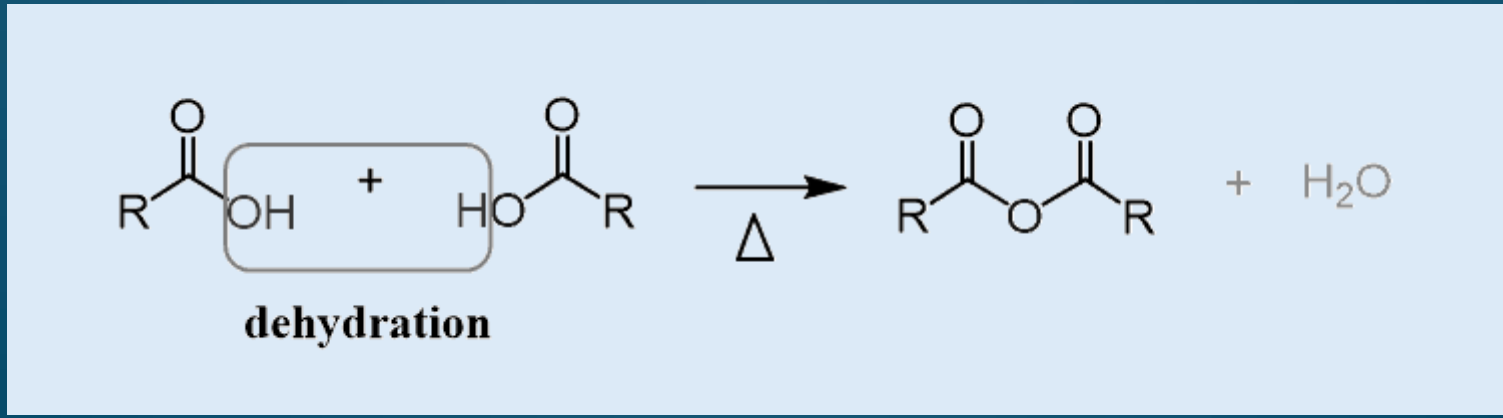
Esters: Saponification

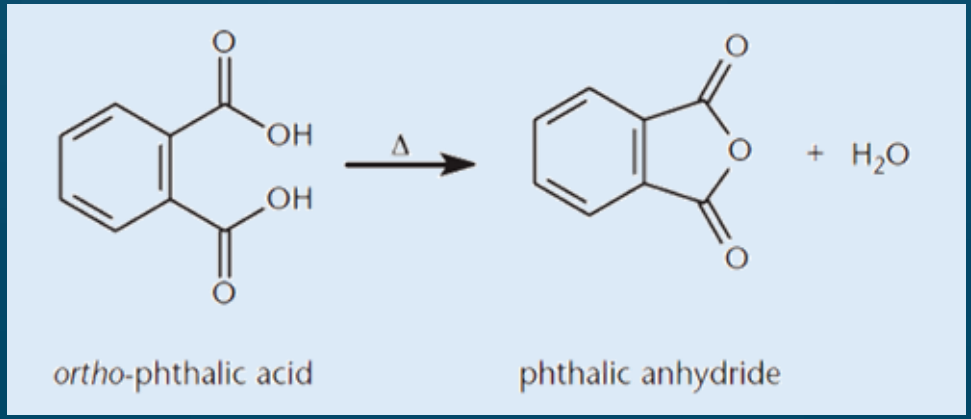


Anhydrides: Nomenclature & Properties



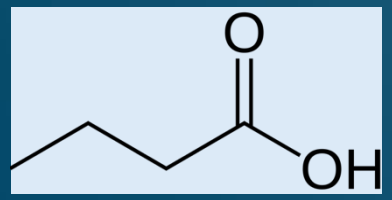
Anhydrides: Formation

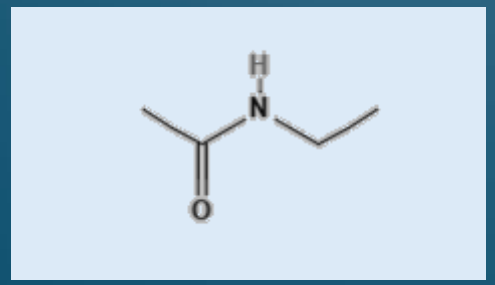


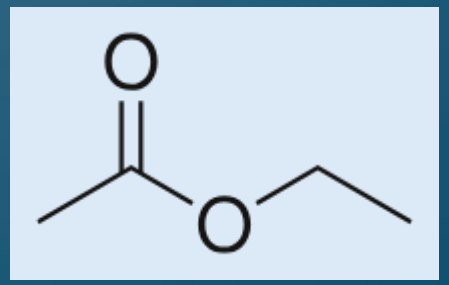


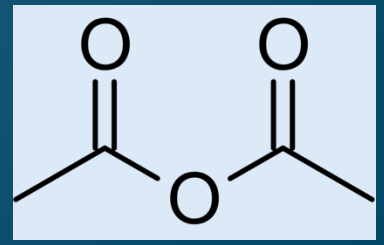
Question

Name each of these compounds and rank them in order of increasing boiling point.



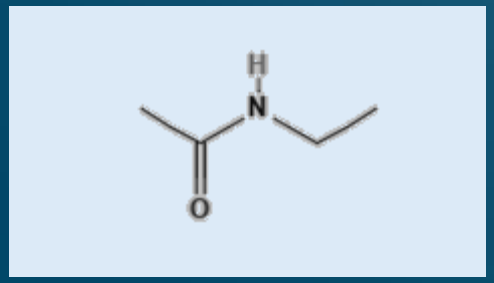


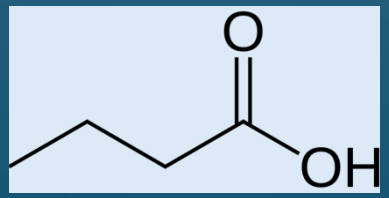


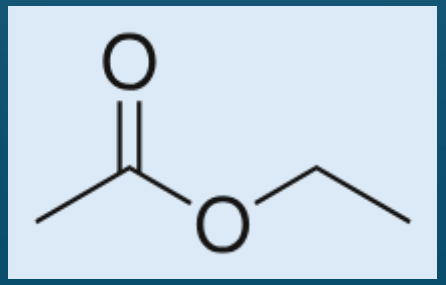


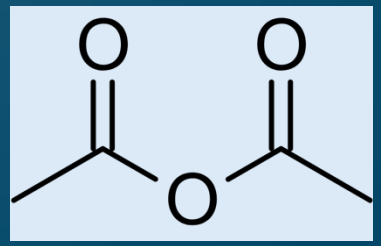
A. I, ii, iii, iv
B. iii, iv, ii, I
C. iii, iv, I, ii
D. iv, iii, ii, i

Reactivity



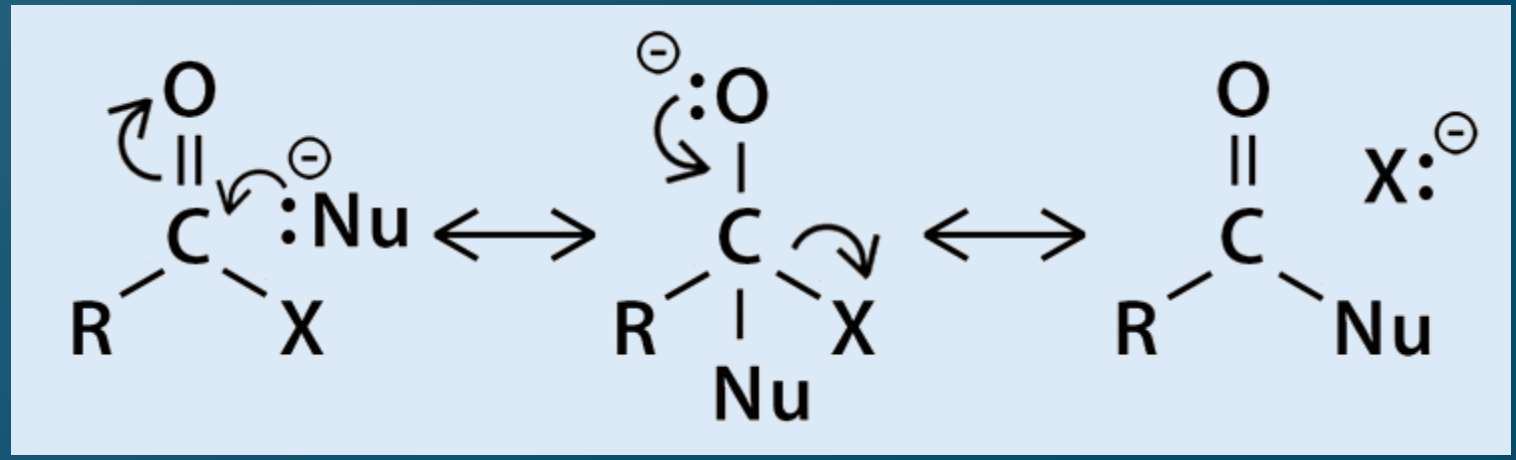




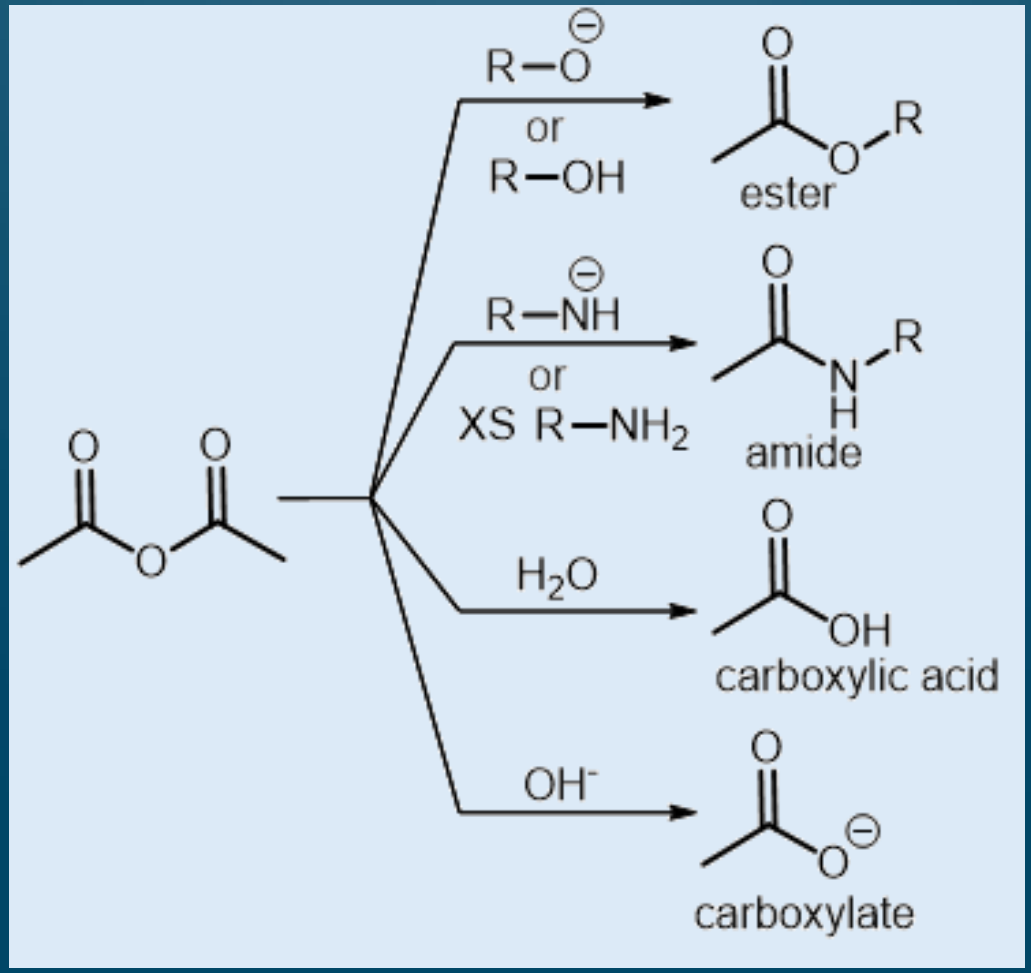


Also consider: ring strain, resonance/conjugation, induction, sterics

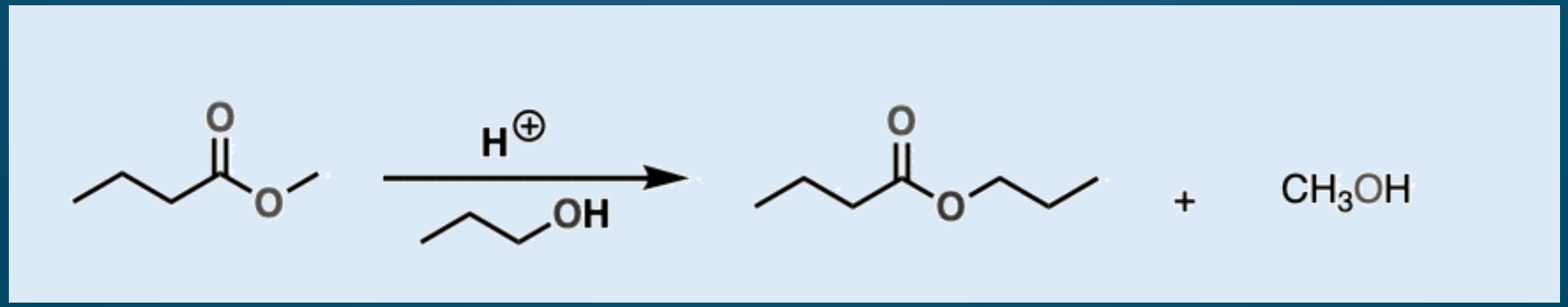
9.3. Nucleophilic Acyl Substitutions



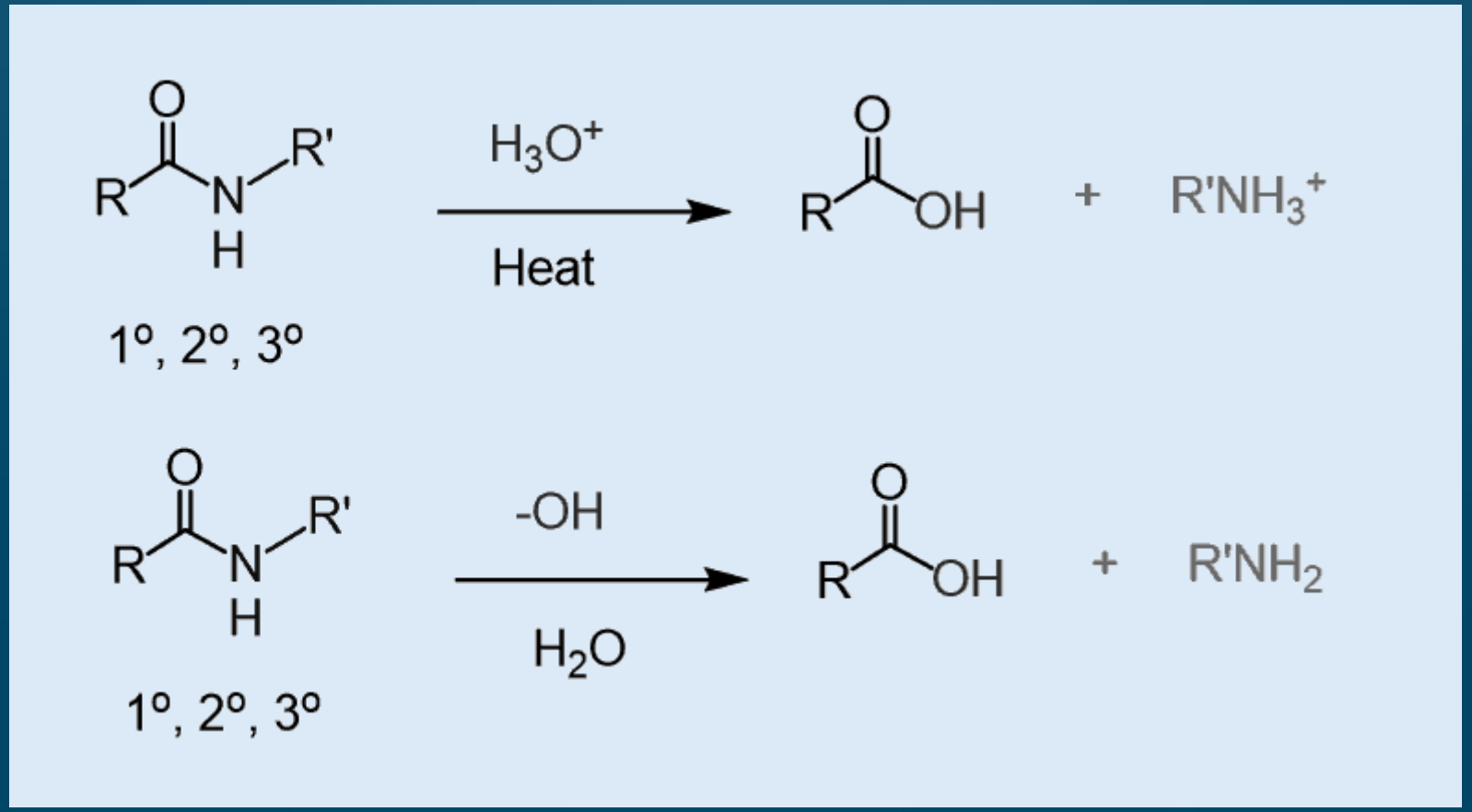
Anhydride Cleavage



Transesterification



Amide Hydrolysis



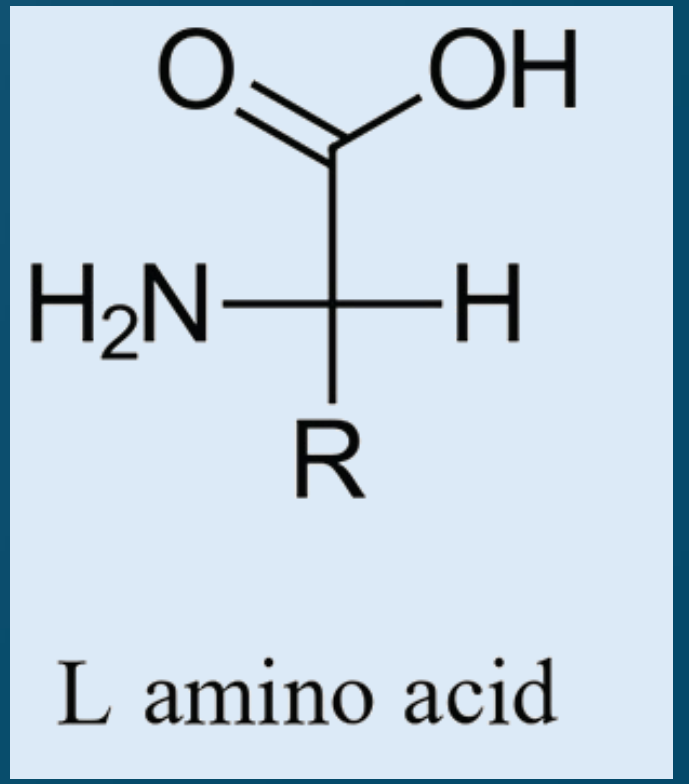
Organic Chemistry Chapter 10

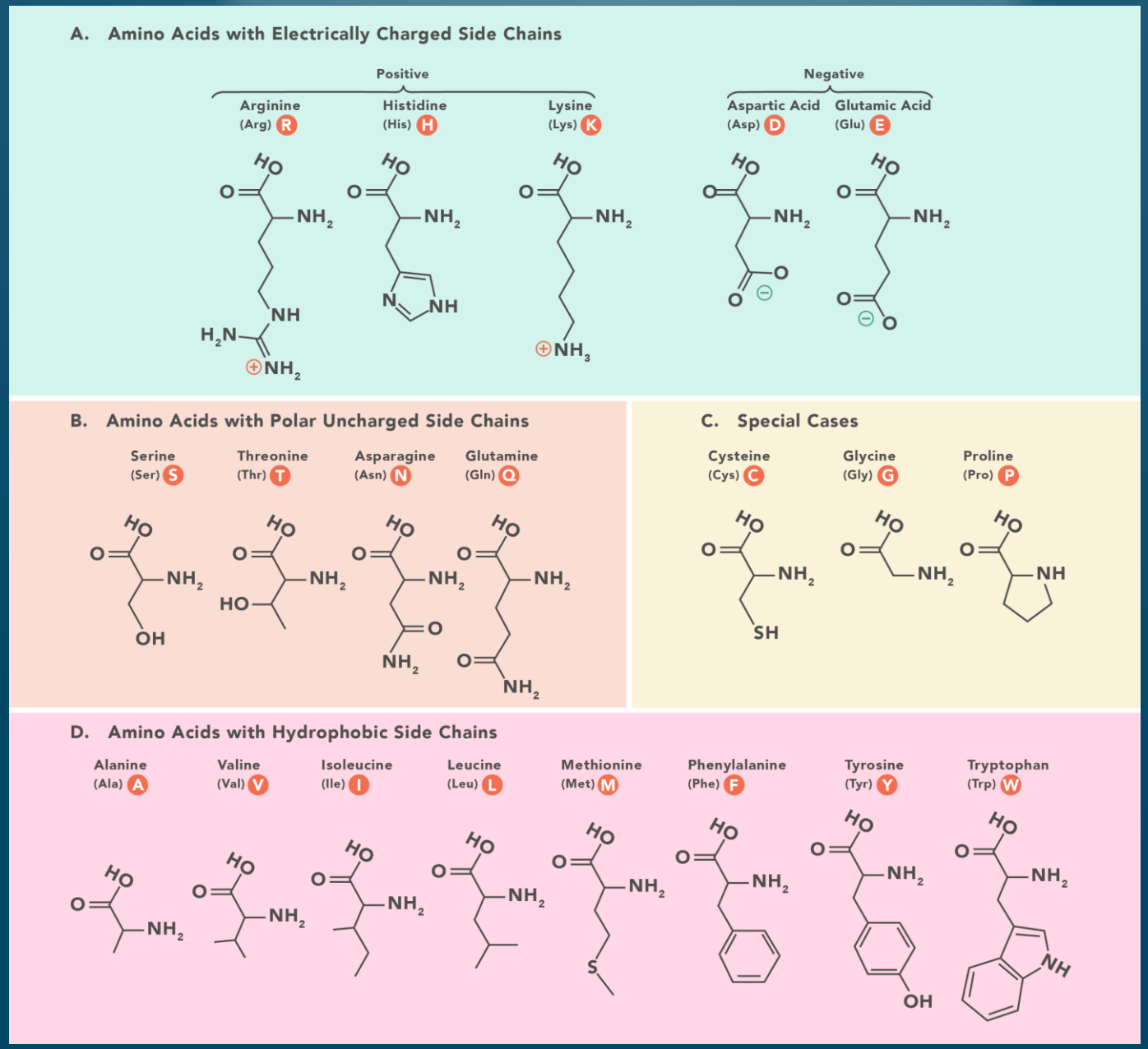
N and P-Containing Compounds



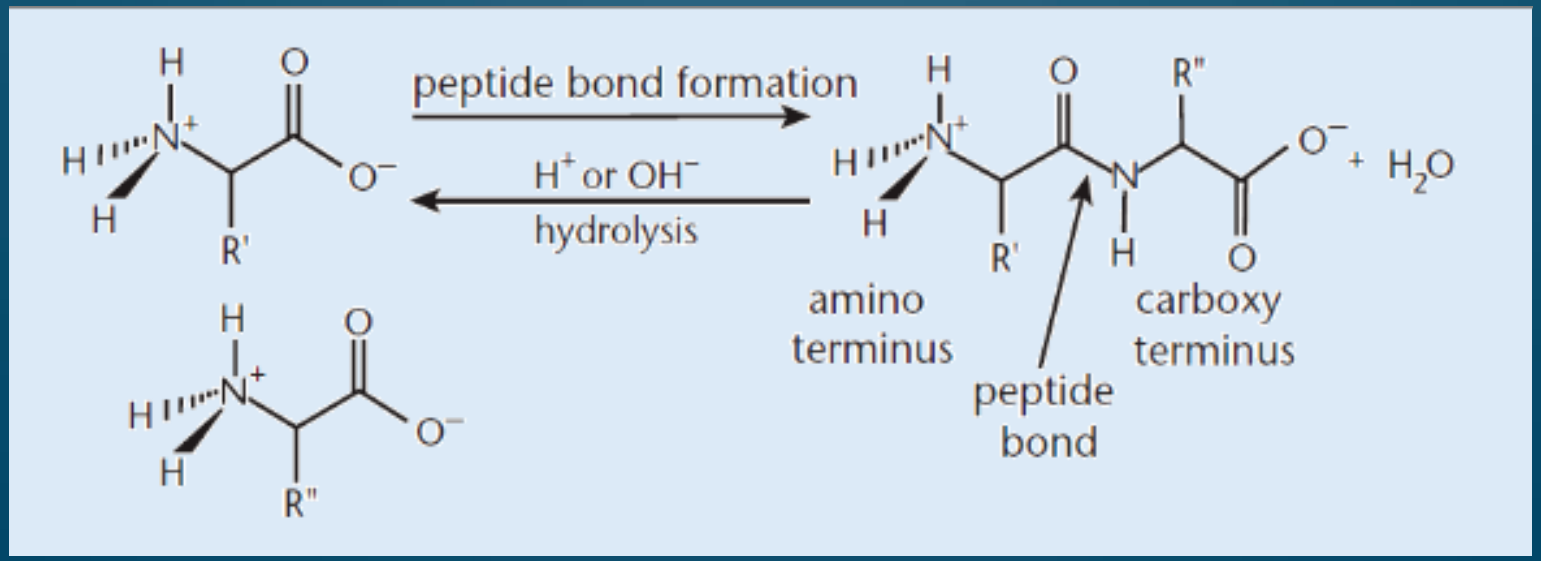
10.1. Amino Acids

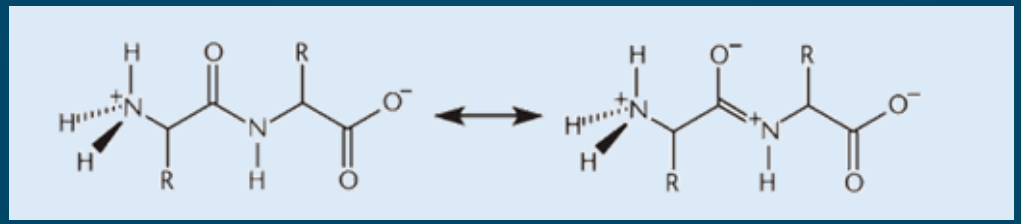





A. Amino Acids with Electrically Charged Side Chains
Positive
Arginine (Arg) R
Histidine (His) H
Lysine (Lys) K
Negative
Aspartic Acid (Asp) D
Glutamic Acid (Glu) E
B. Amino Acids with Polar Uncharged Side Chains
Serine (Ser) S
Threonine (Thr) T
Asparagine (Asn) N
Glutamine (Gln) Q
C. Special Cases
Cysteine (Cys) C
Glycine (Gly) G
Proline (Pro) P
D. Amino Acids with Hydrophobic Side Chains
Alanine (Ala) A
Valine (Val) V
Isoleucine (Ile) I
Leucine (Leu) L
Methionine (Met) M
Phenylalanine (Phe) F
Tyrosine (Tyr) Y
Tryptophan (Trp) W

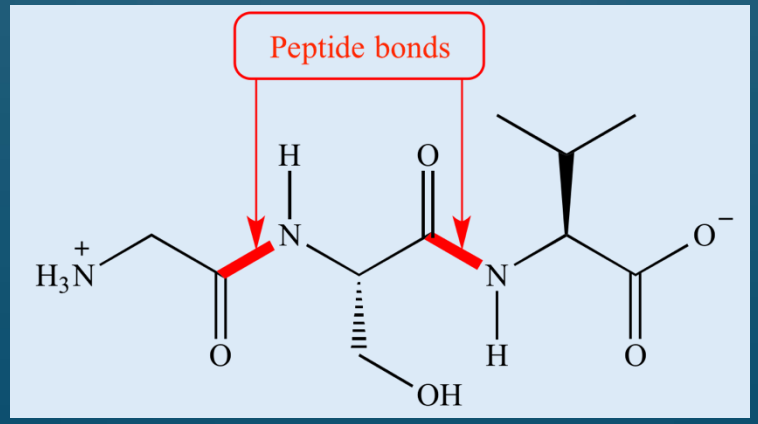
Peptide Bond





Question

What is the hybridization of the peptide bond's N, and what does this indicate regarding its ability to rotate?

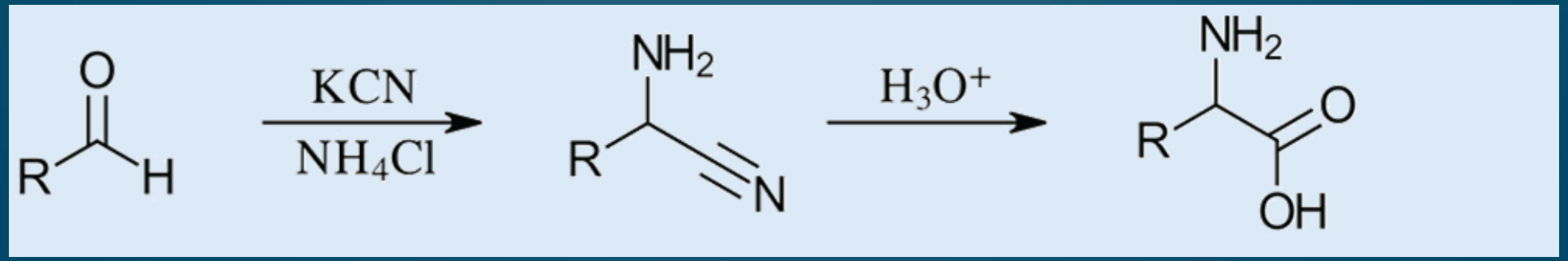


A. sp^3; rotates
B. sp^3; does not rotate
C. sp^2; rotates
D. sp^2; does not rotate

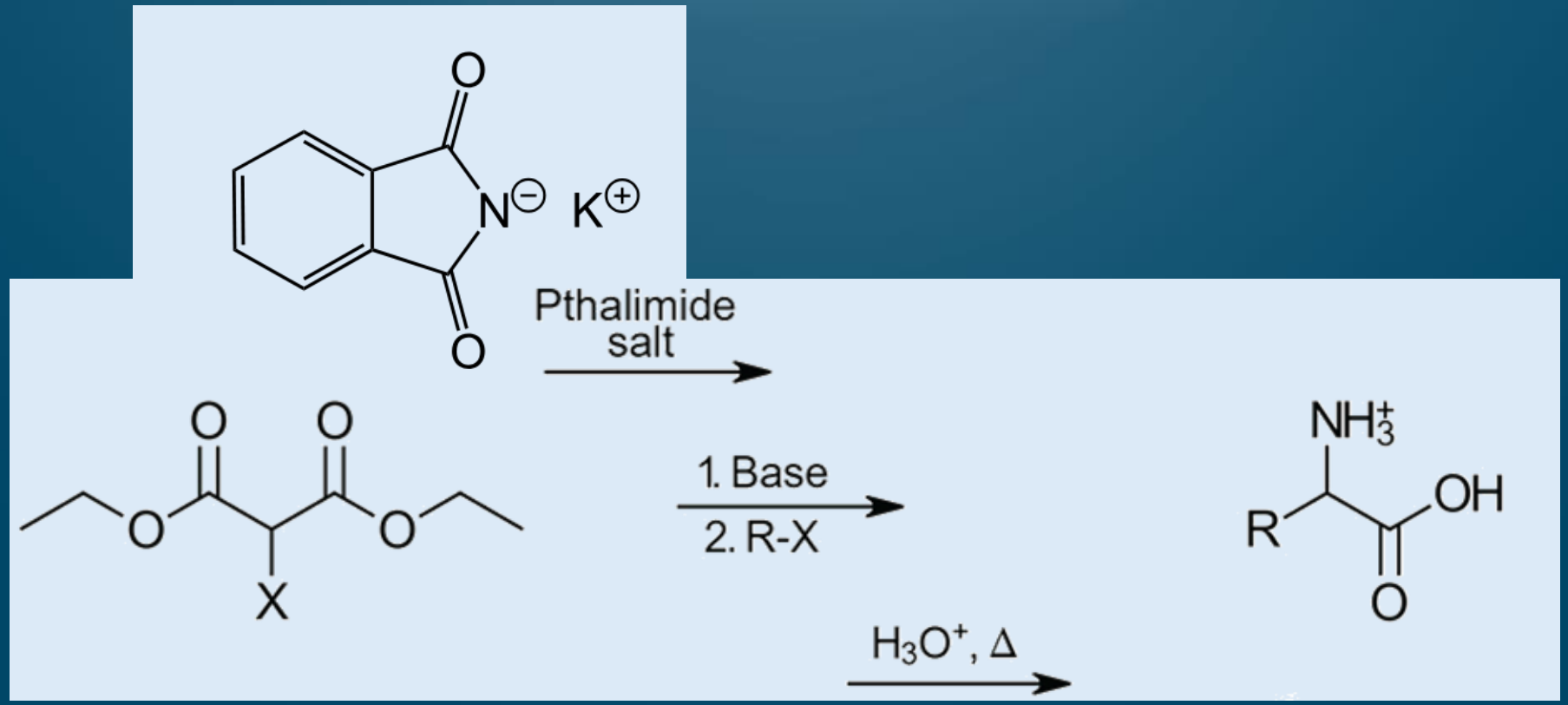
10.2. Amino Acid Synthesis



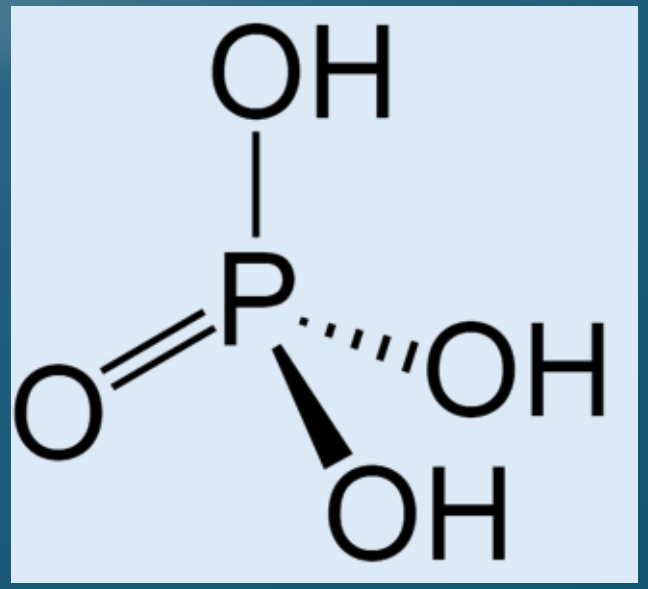
Strecker Synthesis

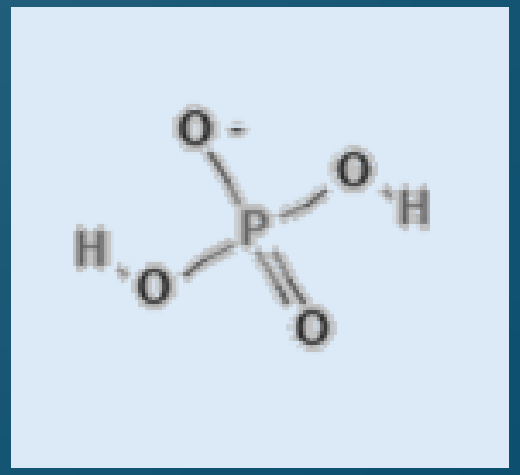


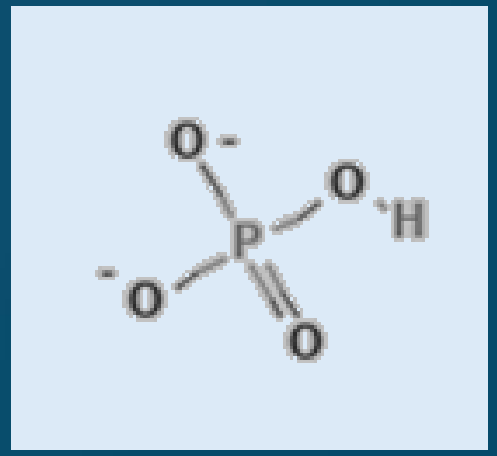
Gabriel Synthesis

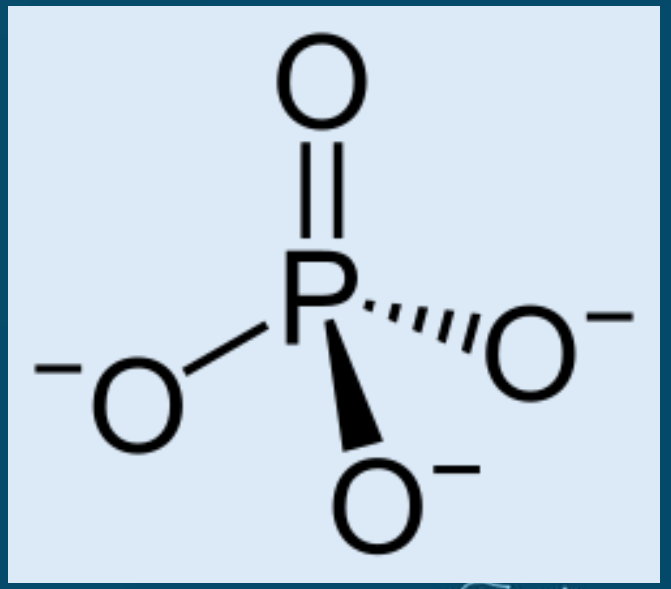


10.3. Phosphorus-Containing Compounds



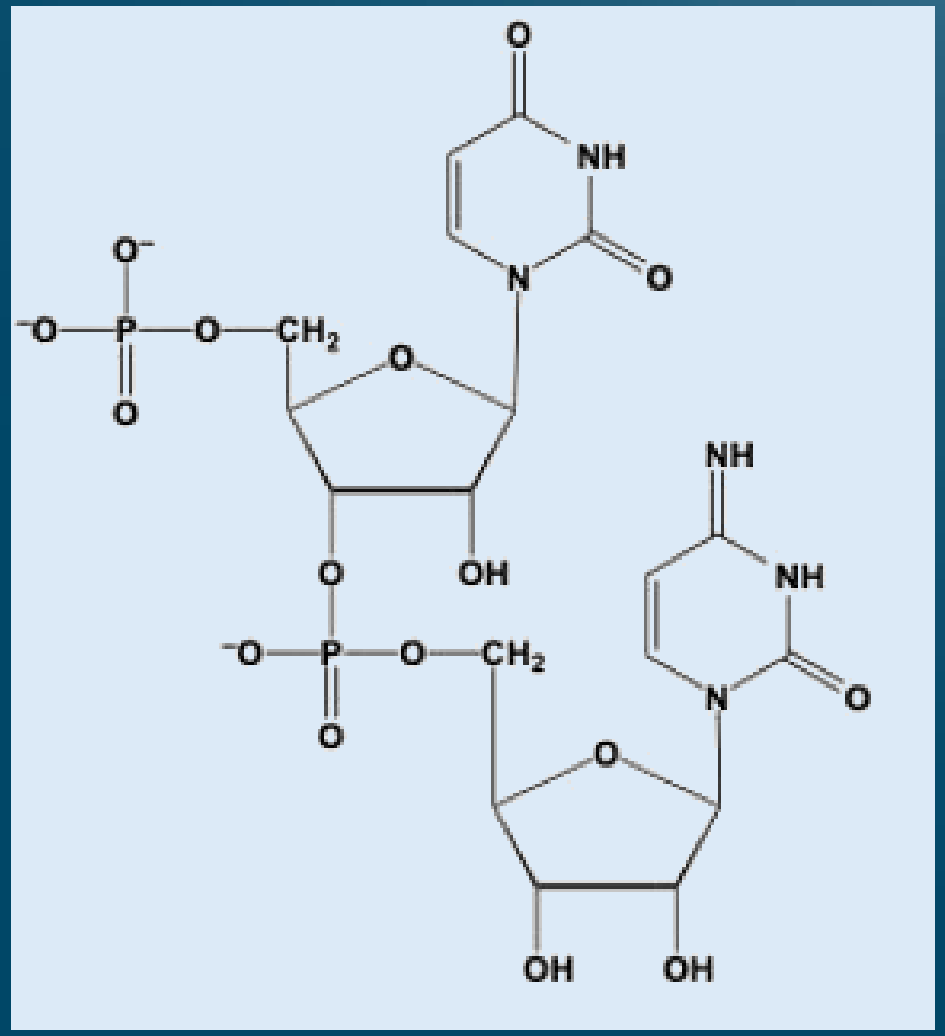


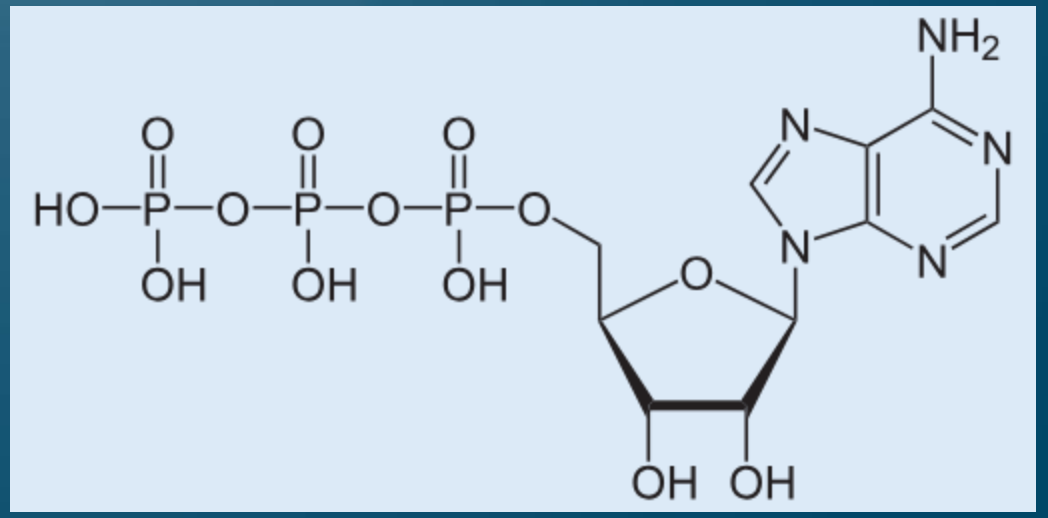


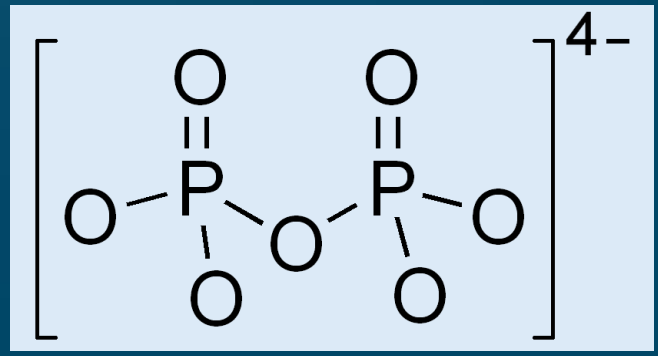


pKa: 2.15 7.20 12.32

Important Molecules







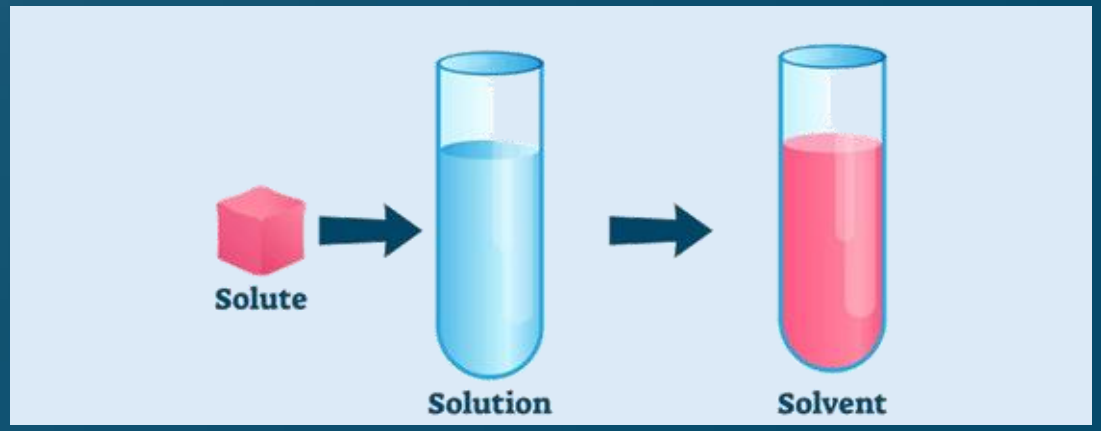
Organic Chemistry Chapter 10

Separation & Purification



12.1. Solubility

$$K_{sp} = [A^+]^x[B^-]^y$$



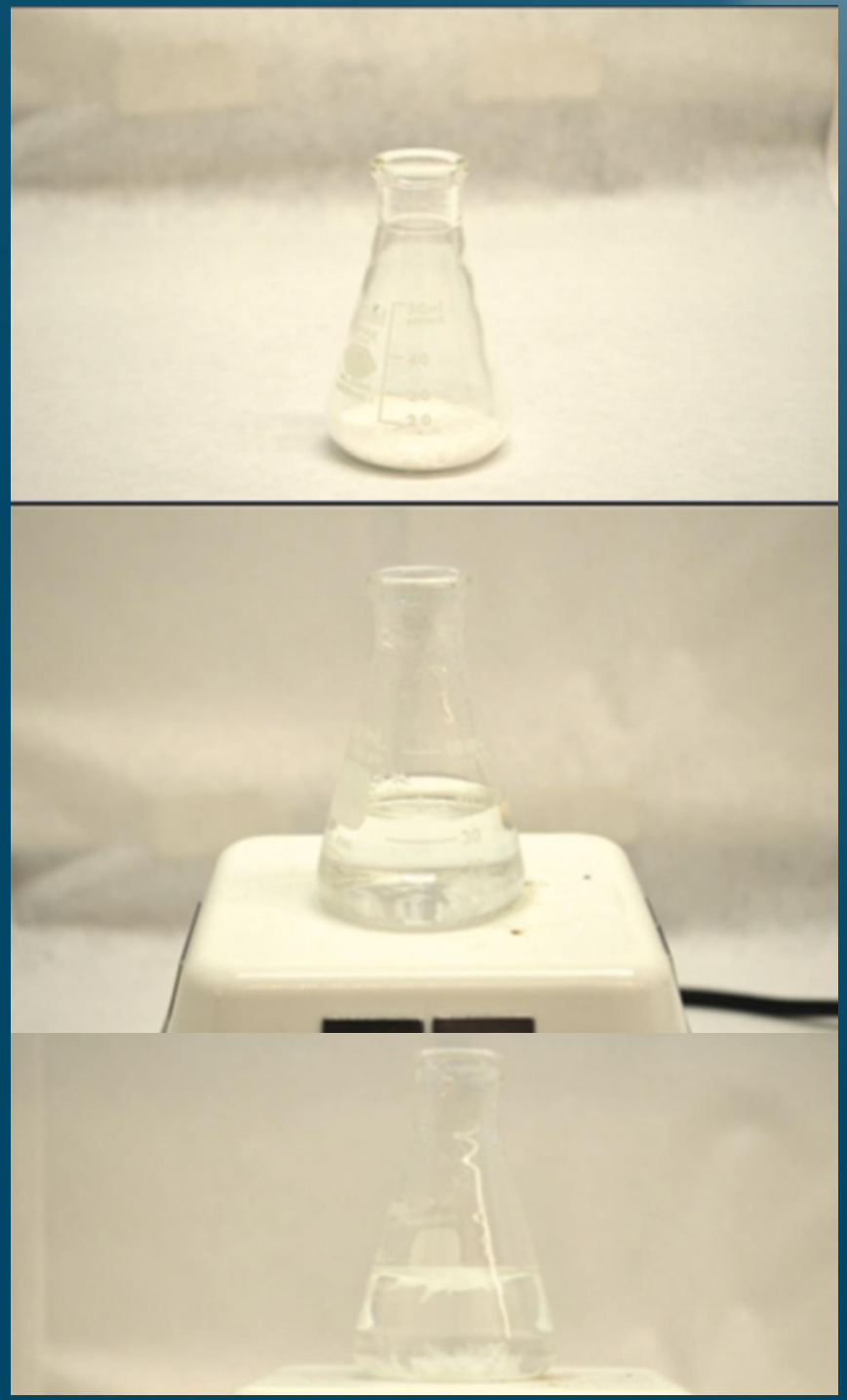
Extraction



Filtration







Recrystallization



Generally Helpful Techniques



Wash out impurities or trace analyte

Question

A student aims to carry out the following reaction:

$$(C_2H_5)_3N(l) + C_2H_5I(l) \rightarrow [(C_2H_5)_4N]^+I^-(s)$$

This reaction takes place in a solvent that both reactants are highly soluble in, but that the product is insoluble in. Which of these solvents would be ideal? (2 acceptable answers)

A. Diethyl ether
B. Toluene
C. Water
D. Acetone



Question

A student aims to carry out the following reaction:

$$(C_2H_5)_3N(l) + C_2H_5I(l) \rightarrow [(C_2H_5)_4N]^+I^-(s)$$

The reactants are somewhat water soluble; in a nonpolar solvent, the product will precipitate as large white crystals. What equipment should be used to extract product?

A. Buchner funnel w/vacuum
B. Buchner funnel w/o vacuum
C. Separatory funnel
D. Rotovap



Question

A student aims to carry out the following reaction:

$$(C_2H_5)_3N(l) + C_2H_5I(l) \rightarrow [(C_2H_5)_4N]^+I^-(s)$$

A two-step washing process is carried out in order to first wash the product, then redissolve It (after filtrate has been discarded). Which two solvents, in order, would be ideal for this process?

A. Toluene, hexane
B. Hexane, water
C. Base, water
D. Water, acid



12.2. Distillation

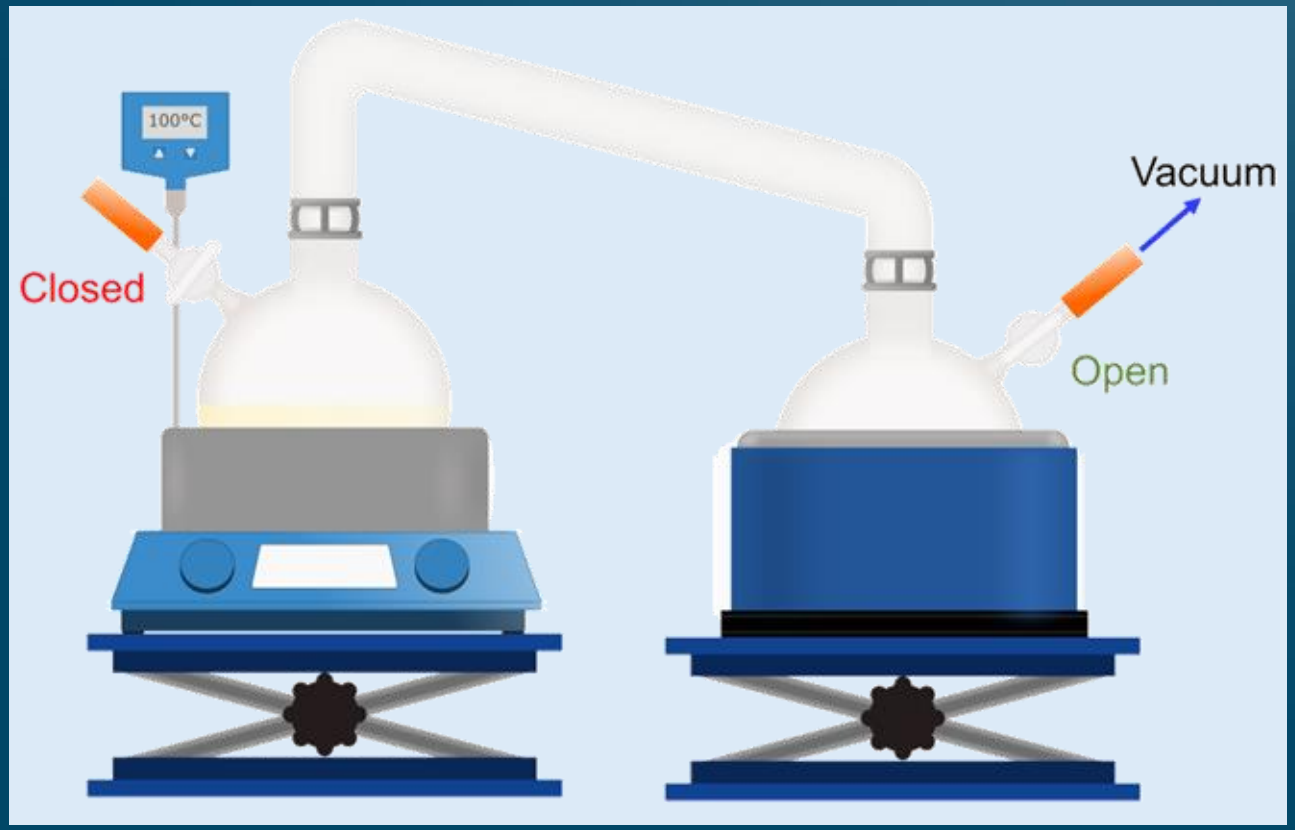


Simple Distillation



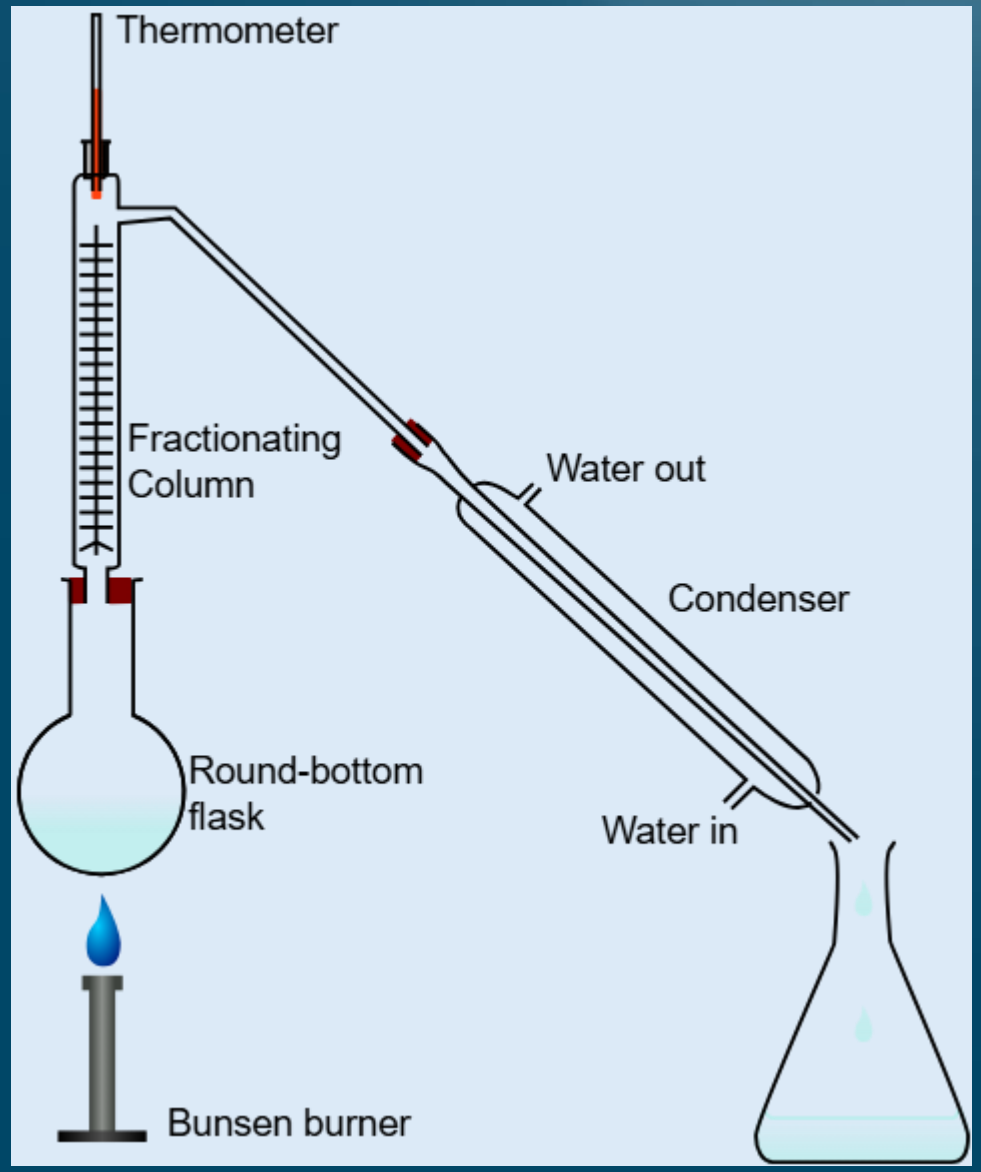
-b.p. difference of >25C
-b.p. <150C

Vacuum Distillation



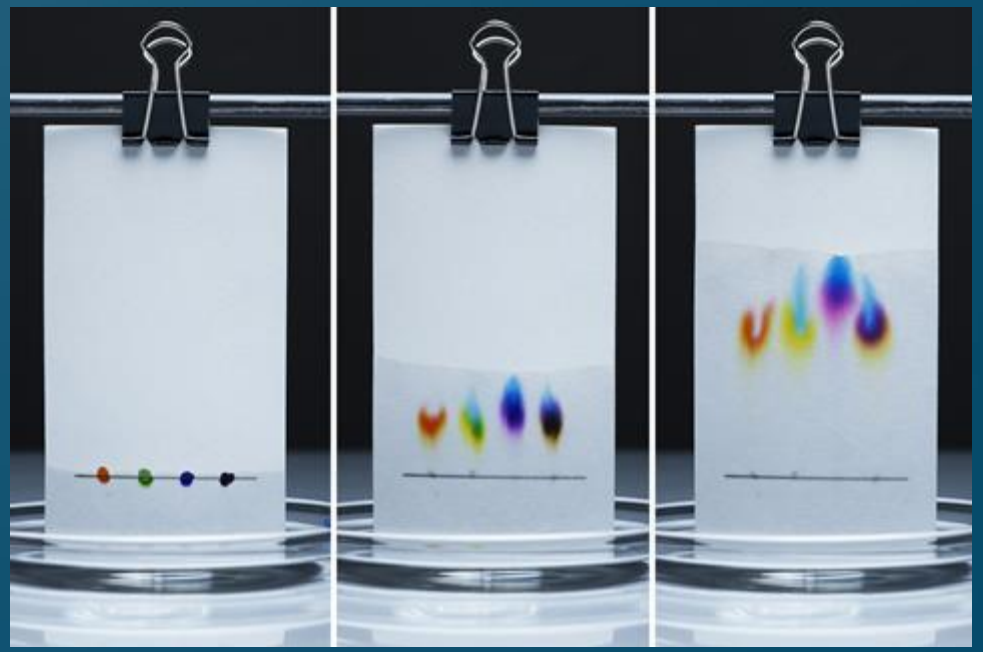
-b.p. >150C
-vacuum

Fractional Distillation

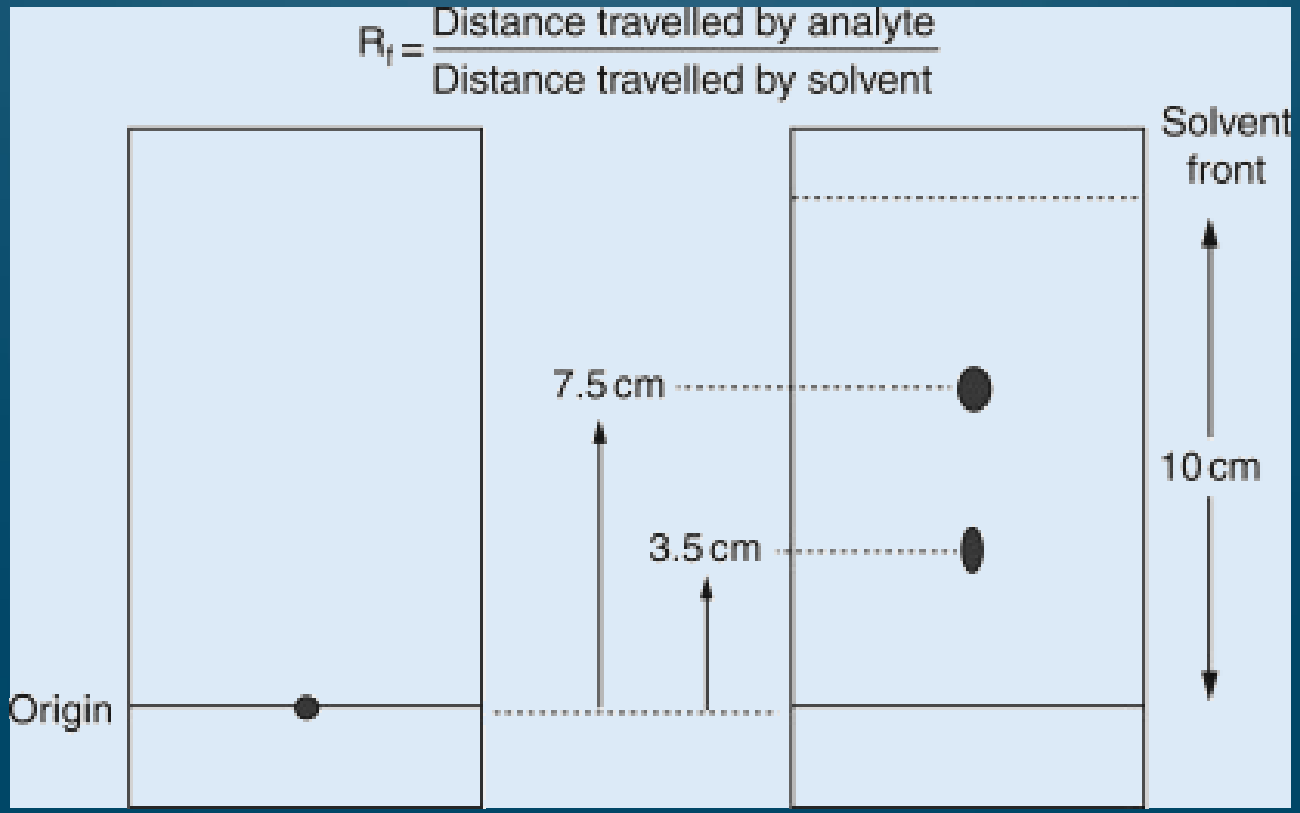


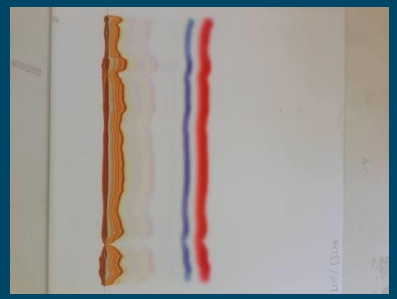
-b.p. difference of <25C
-fractionating column

12.3. Chromatography

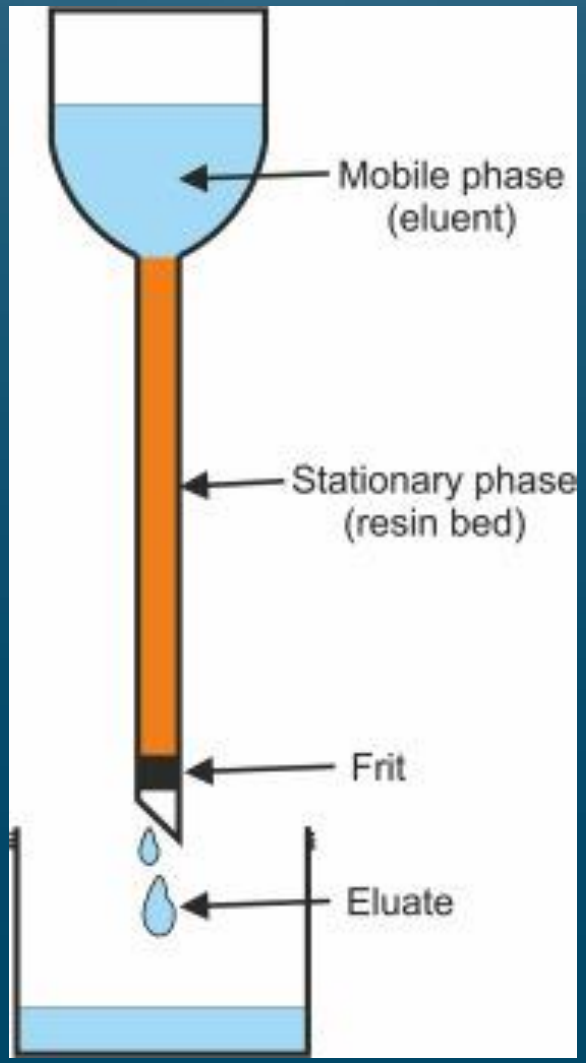


Thin Layer Chromatography (TLC)

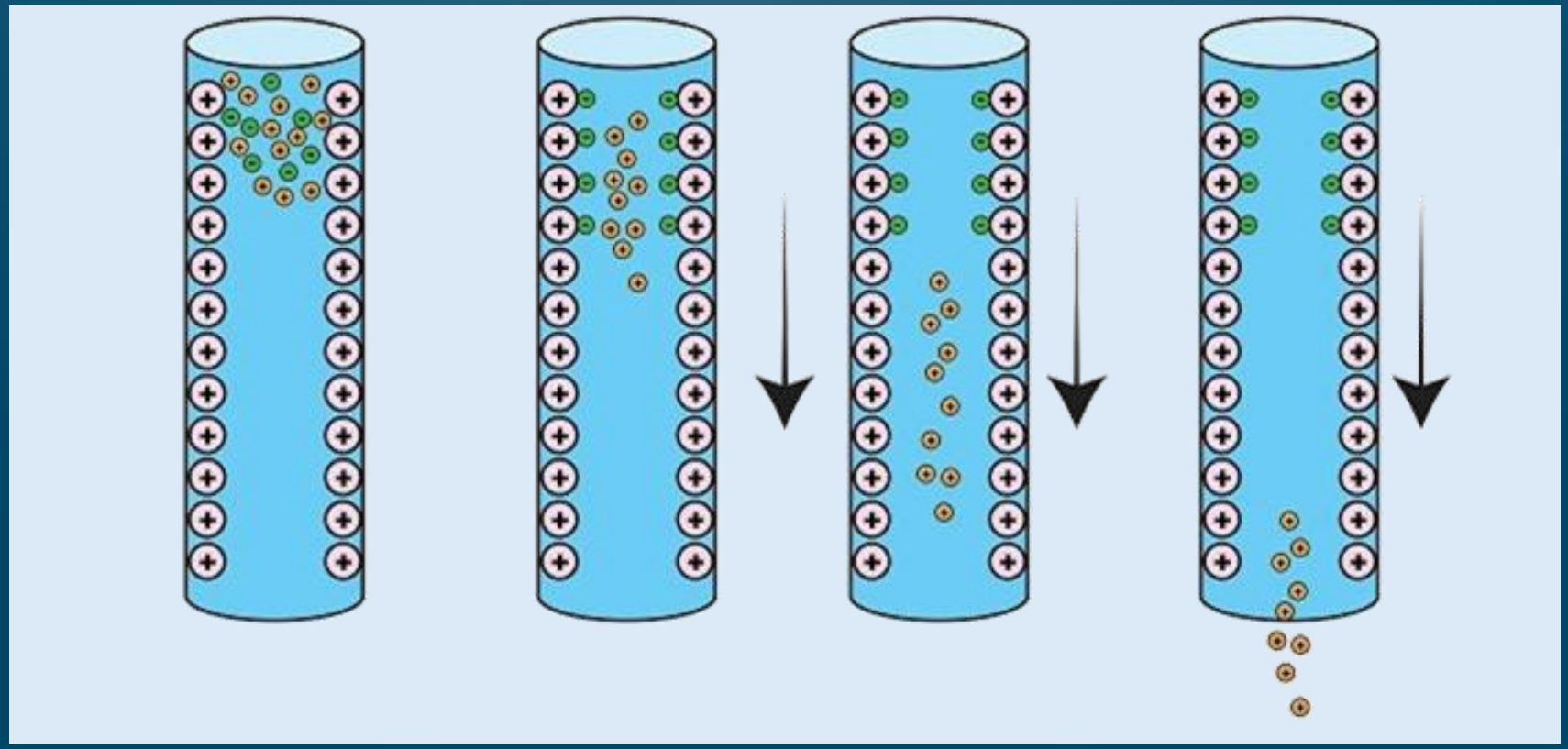




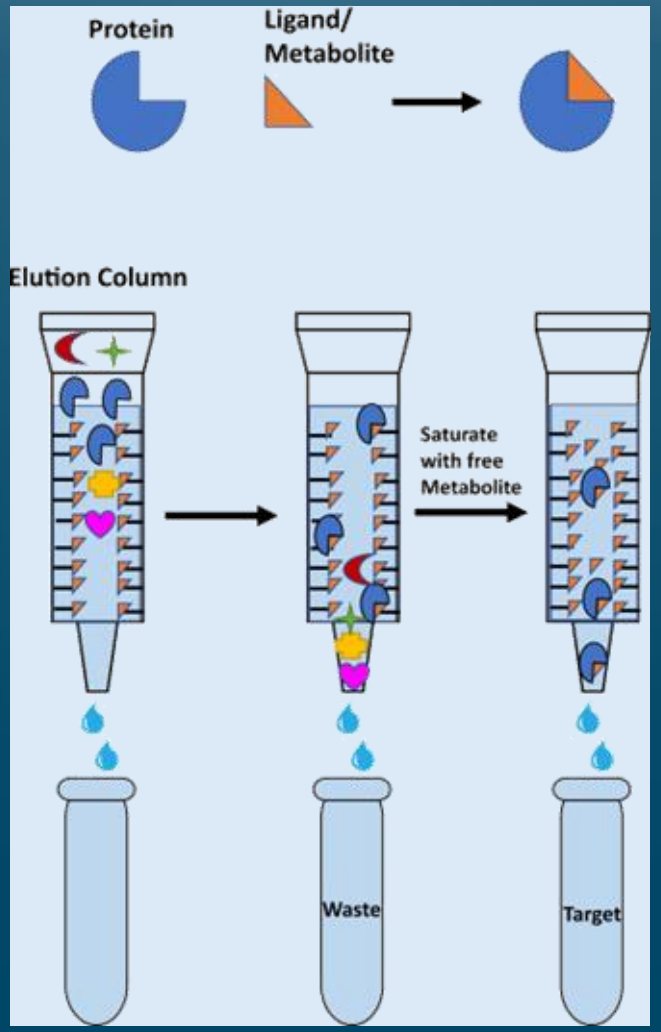
Column Chromatography



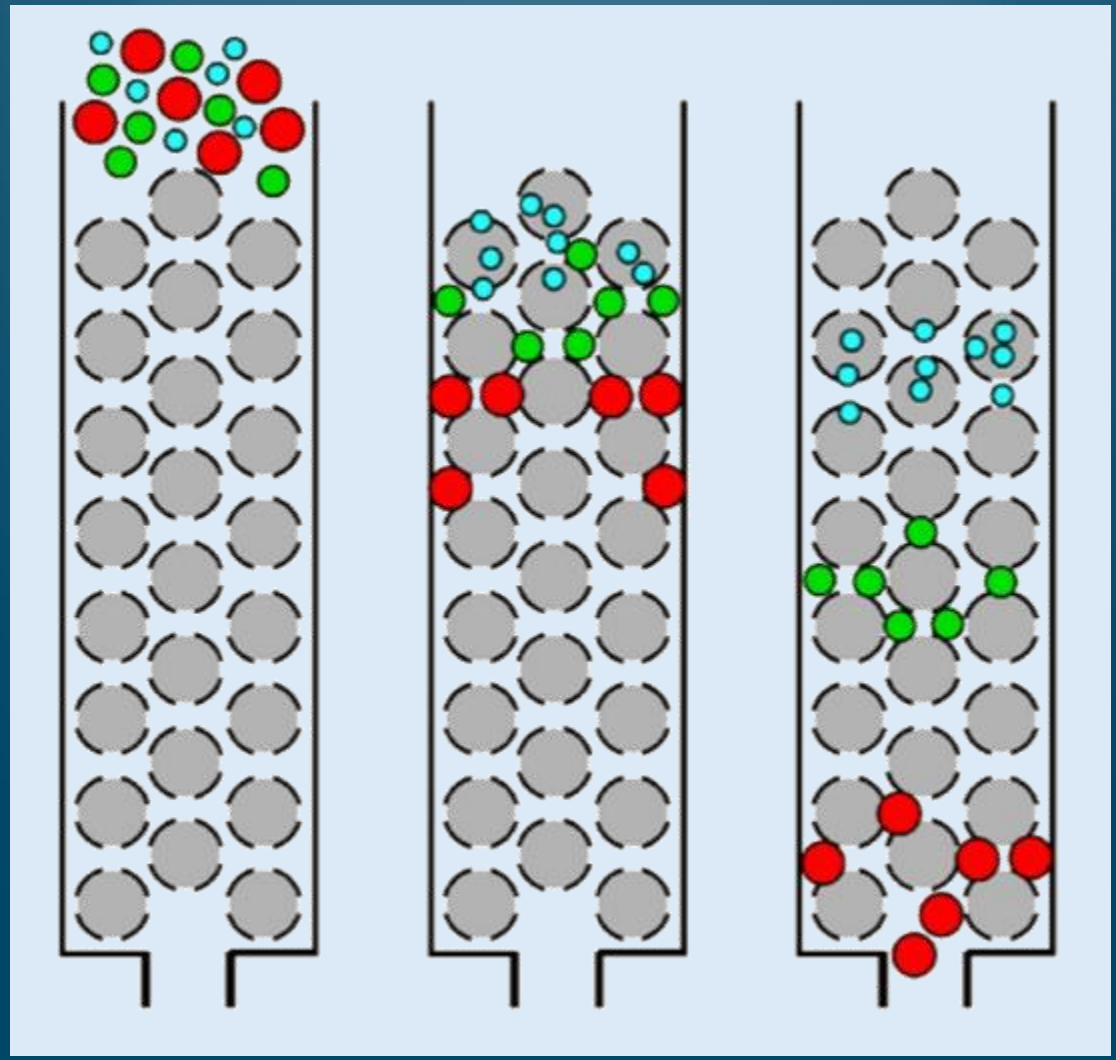
Ion-Exchange Chromatography



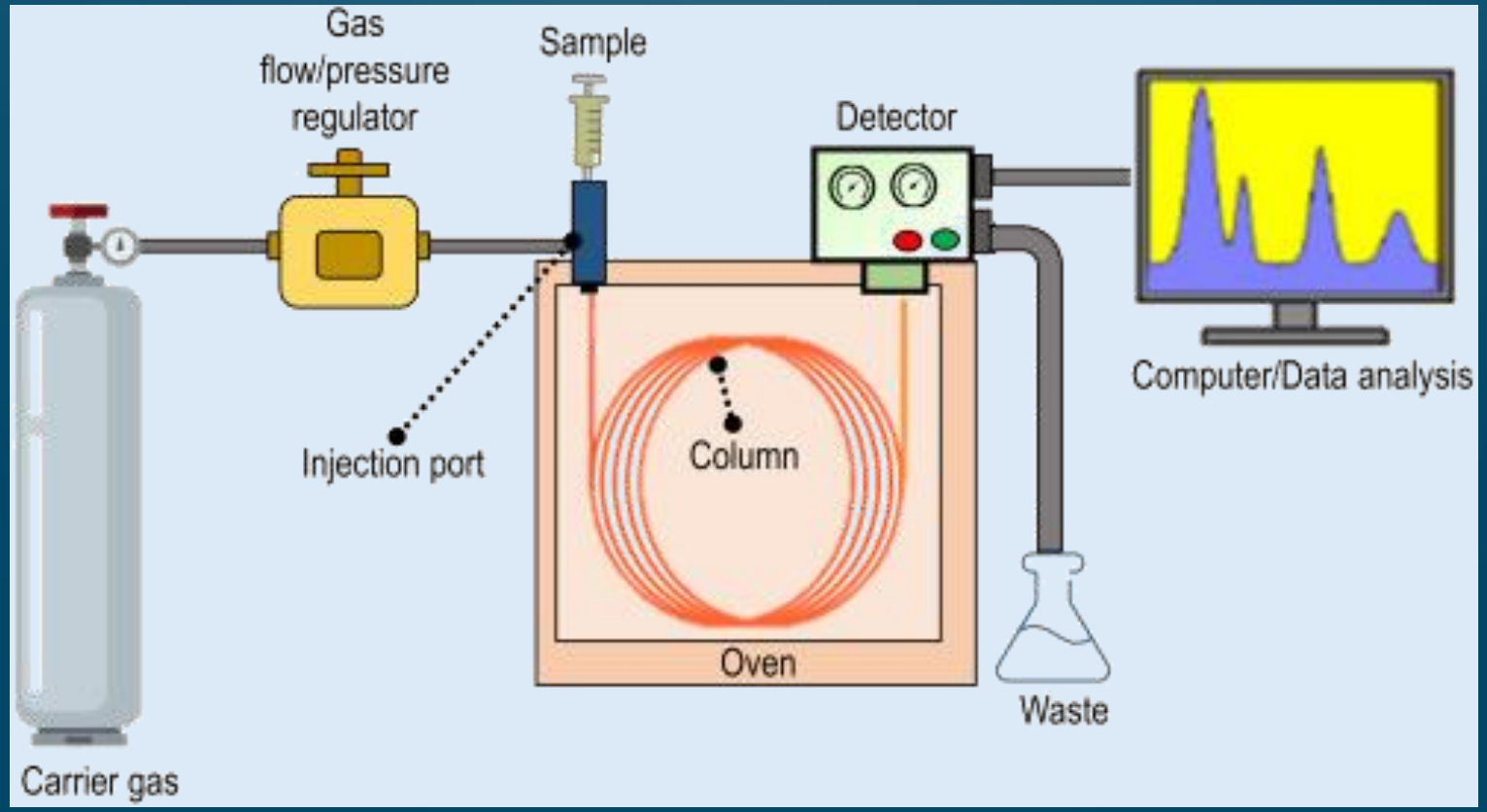
Affinity Chromatography



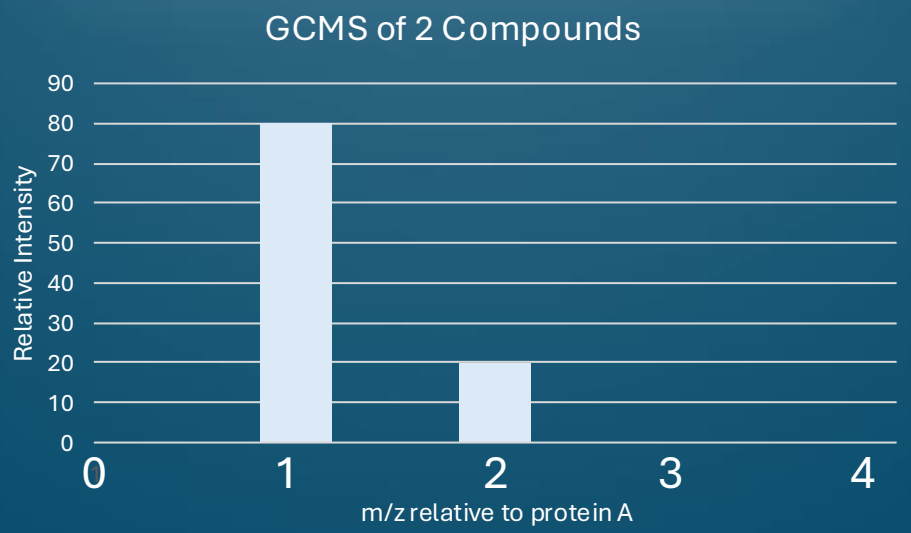
Size Exclusion Chromatography



Gas Chromatography



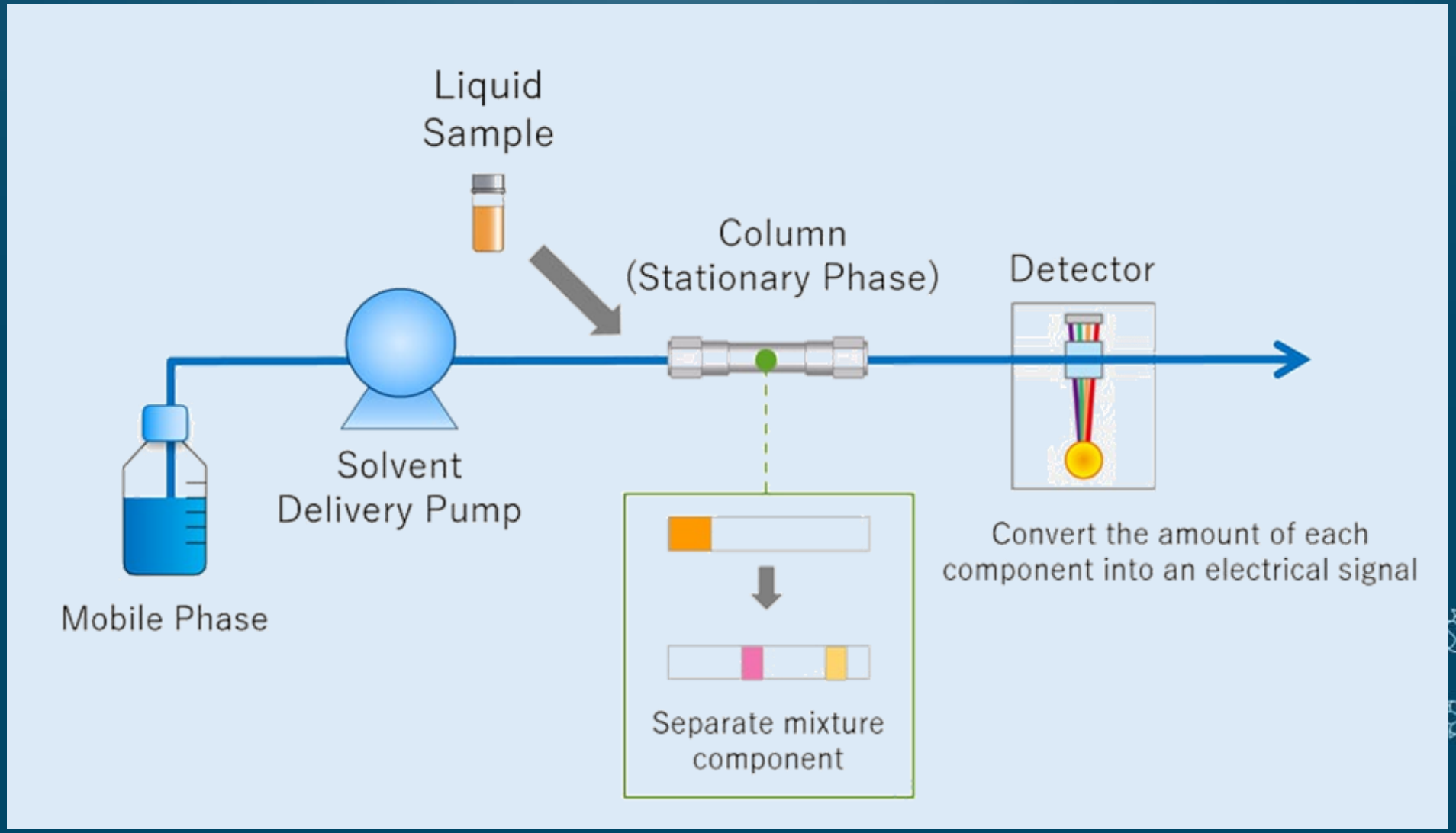
Question



Two proteins are analyzed using GCMS. Protein A is most abundant and protein B is present in smaller quantities. Let protein A be 150 Da with an overall charge of +4. Which of these masses + charges matches those of protein B? Select all that apply.

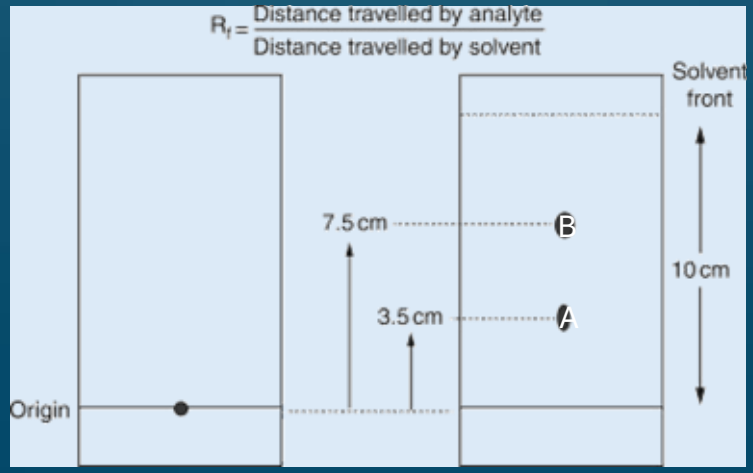
A. 150 Da, no charge
B. 300 Da, +4
C. 75 Da, +2
D. 150 Da, +2

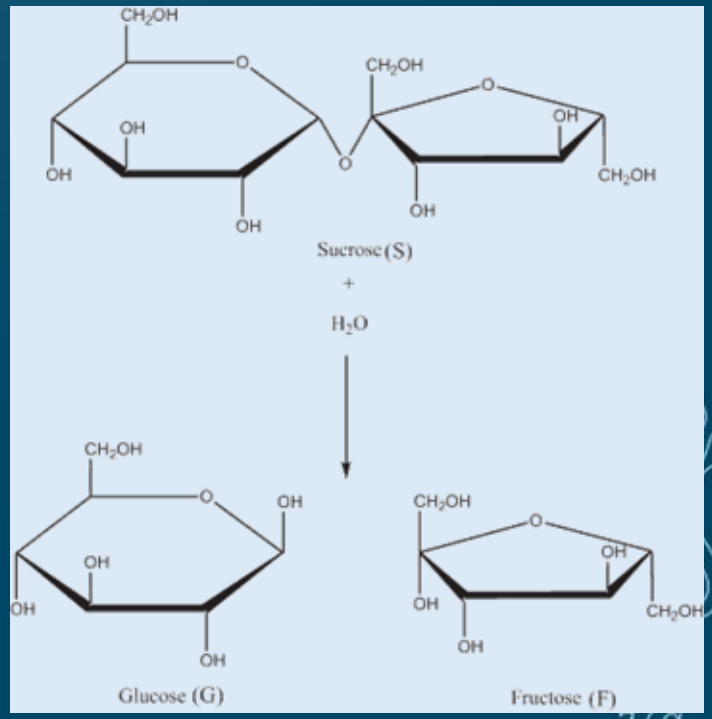
HPLC


Liquid
Sample
Column
(Stationary Phase)
Detector
Solvent
Delivery Pump
Mobile Phase
Separate mixture
component
Convert the amount of each
component into an electrical signal

Question

Sucrose is a disaccharide composed of glucose and fructose linked through a glycosidic bond. Under acidic conditions, sucrose can undergo hydrolysis to produce glucose and fructose.
A researcher investigated the progress of this reaction using thin-layer chromatography (TLC). A cellulose plate was used as the stationary phase, while a relatively nonpolar organic solvent mixture served as the mobile phase. The reaction mixture was spotted, with initial and final results pictured. Other analysis methods determined that product yield was determined to be about 60%.



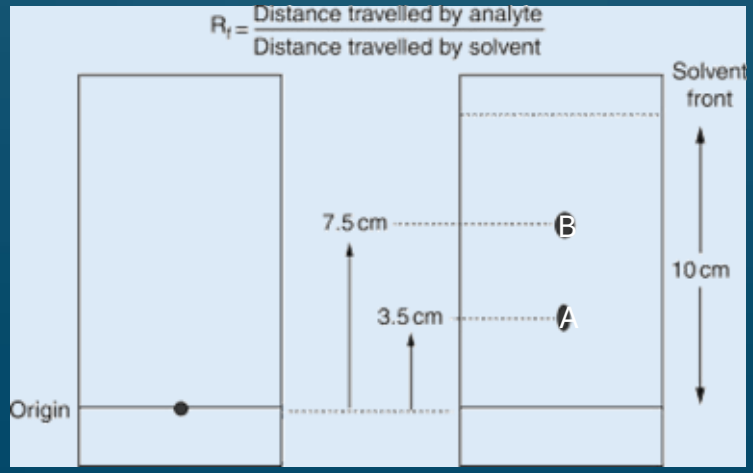


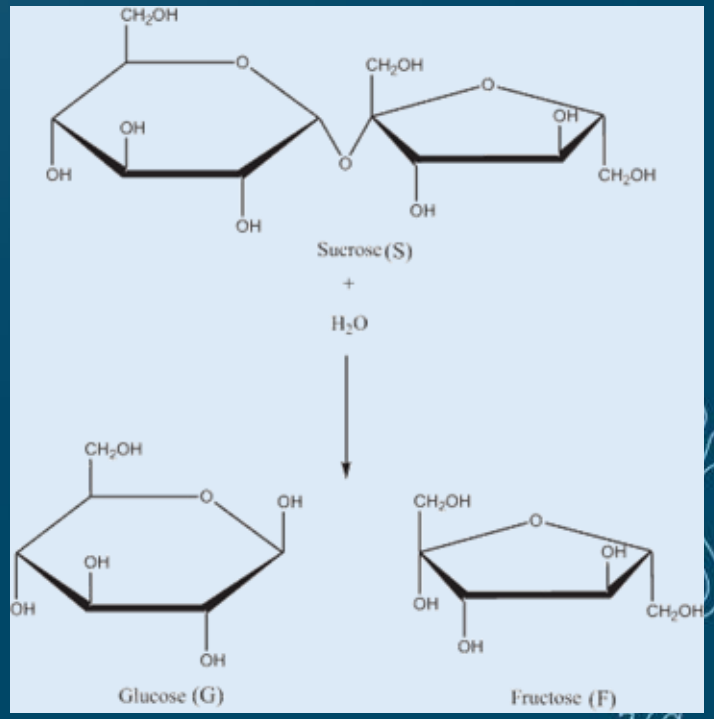
Calculate the Rf of A, then B.

A. 0.035, 0.075
B. 0.5, 1.65
C. 0.075, 0.35
D. 0.35, 0.75

Question

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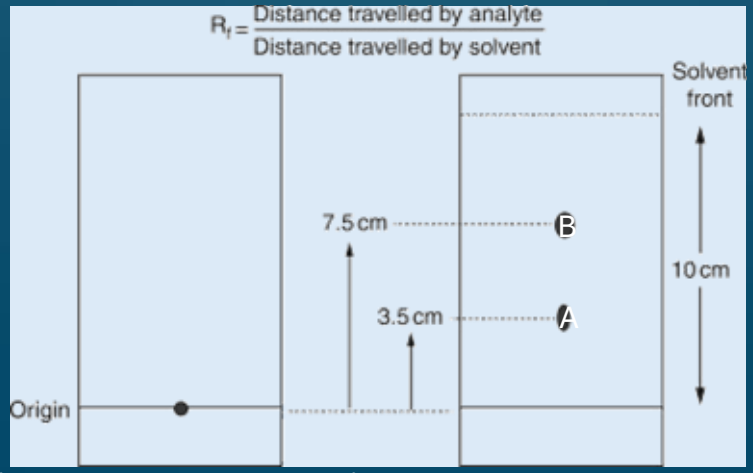


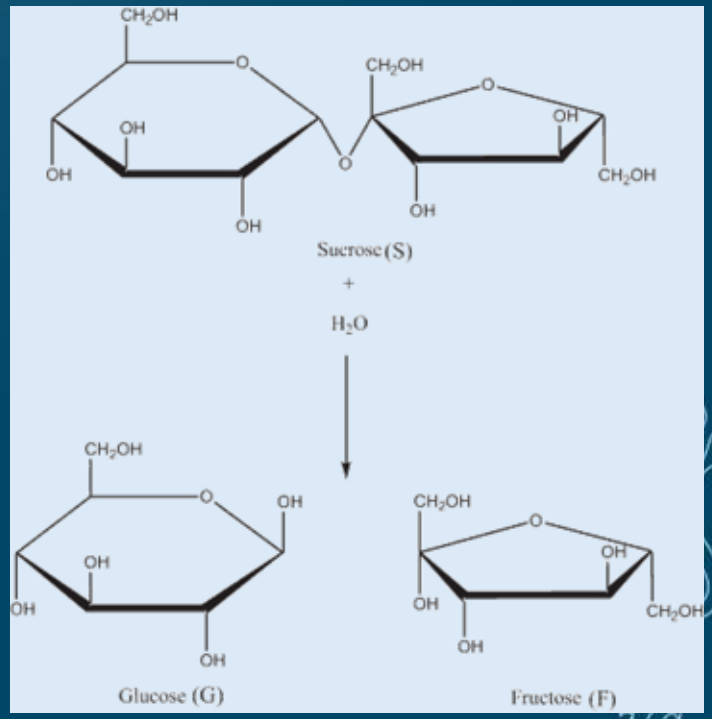
Identify the likely compound(s) in A, then B.

A. Glucose, Fructose
B. Fructose, glucose
C. Sucrose, glucose+fructose
D. Glucose+fructose, sucrose

Question

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What method would likely best partition all three species?

A. Leave TLC plate for longer
B. Ion exchange chromatography
C. HPLC
D. GCMS
E. Size exclusion chromatography